Demographics for Civilization VII

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Figure 1: Demographics logo

Demographics samples every civilization each turn and turns what it records into charts, leaderboards, and diplomacy views. It reads the game and never changes it: no yields, balance, or AI behavior are touched.

At a glance

- **Global Statistics:** per-turn line charts of yields, economy, resources, society, religion, land, and settlements, with every civilization on one plot and history carried across age changes.
- **Geopolitics:** a diplomacy ring of the civilizations and city-states you have met, agreements over time, each crisis broken down stage by stage, soft-power comparisons, and a full military record with a timeline of every war.
- **History:** a chronicle of your campaign, told age by age (settlements founded and taken, wonders, wars and peace, religions, Triumphs, crises, disasters, new ages), an interactive timeline with a territory map that replays your expansion, and a lineage of every leader and the civilization they led in each age.

Figure 2: Gold per turn for every civilization across Antiquity and Exploration, with crisis stages, wonders, and the age change marked on the timeline

- **World Rankings:** civilization and settlement leaderboards, a sortable rank-by-yield table for each, a Top 25 settlements board that can fly the camera to any city, and a **Hall of Fame** that ranks every campaign you have played, also open from the main menu.
- **Real-world scale:** populations in the thousands to tens of millions by era, GDP in dollars, and territory in km².
- **Your view, your way:** per-civ colors, time-range filters, hide and focus, civ-or-leader labels, colorblind mode, CSV export, and a Basic mode that shows only the core pages.
- **Companion-ready:** the *Emigration* mod adds its own tab, and any mod can add metrics or pages through a small public API.

Built on the foundations of robk's InfoAddict (Civ V), Gedemon's CivGraphs (Civ VI), and Slothoth's Global Relations panel (Civ VII). Open source, with full readable source included.

Screenshots

Every view of the dashboard, 64 screenshots in all, is attached to each GitHub release.

System Guide and Feature Reference

Contents

1. Layout
2. Global Statistics
3. Geopolitics
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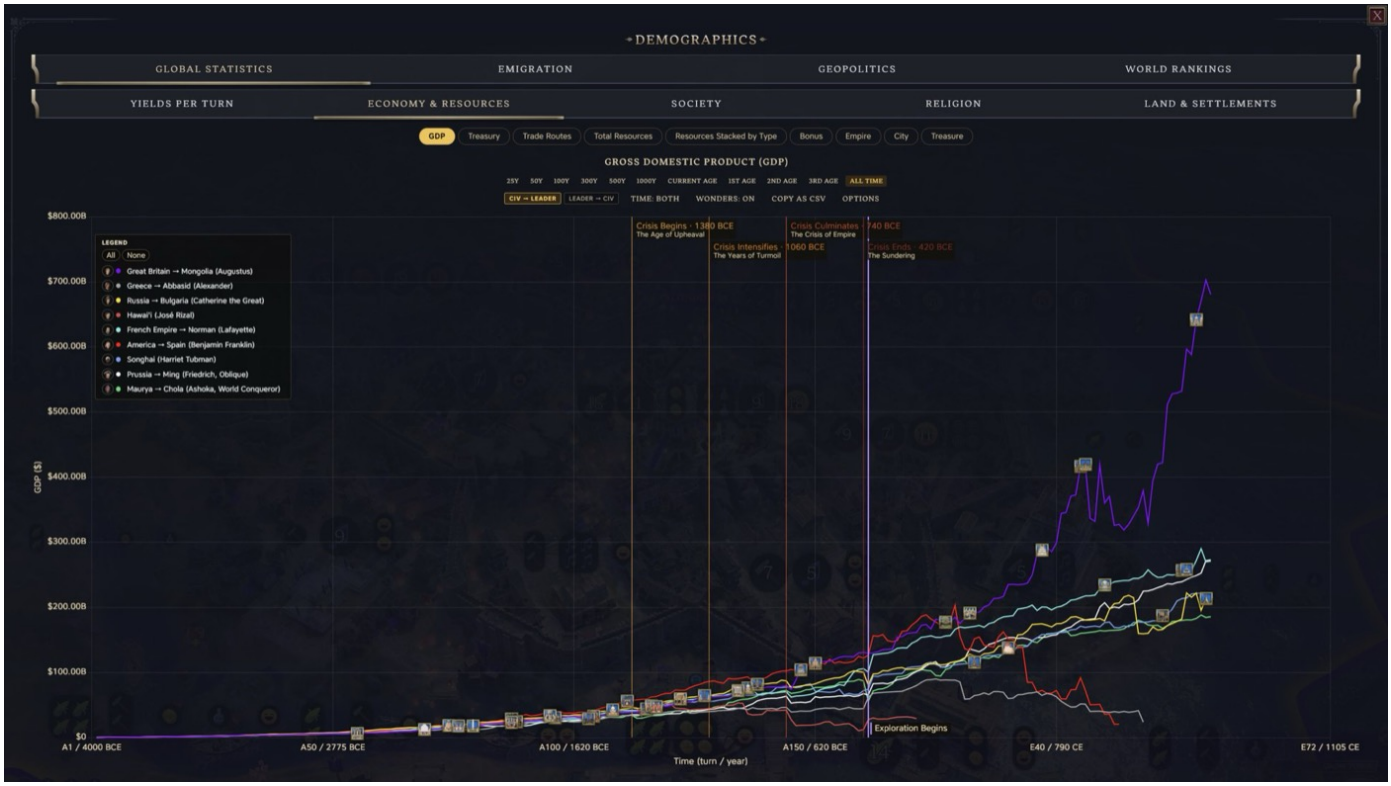

Figure 3: GDP over time, shown in real-world dollars

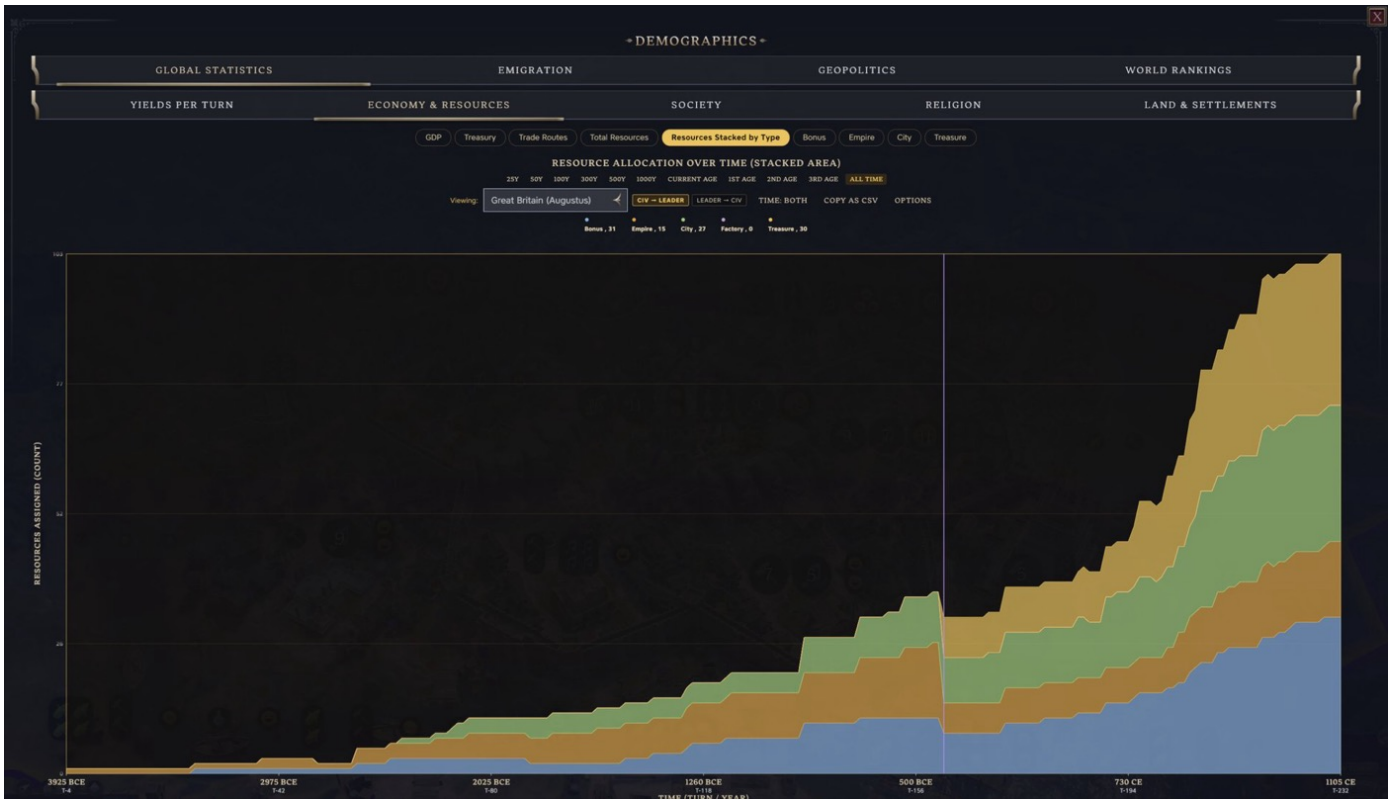

Figure 4: Resources held, stacked by type

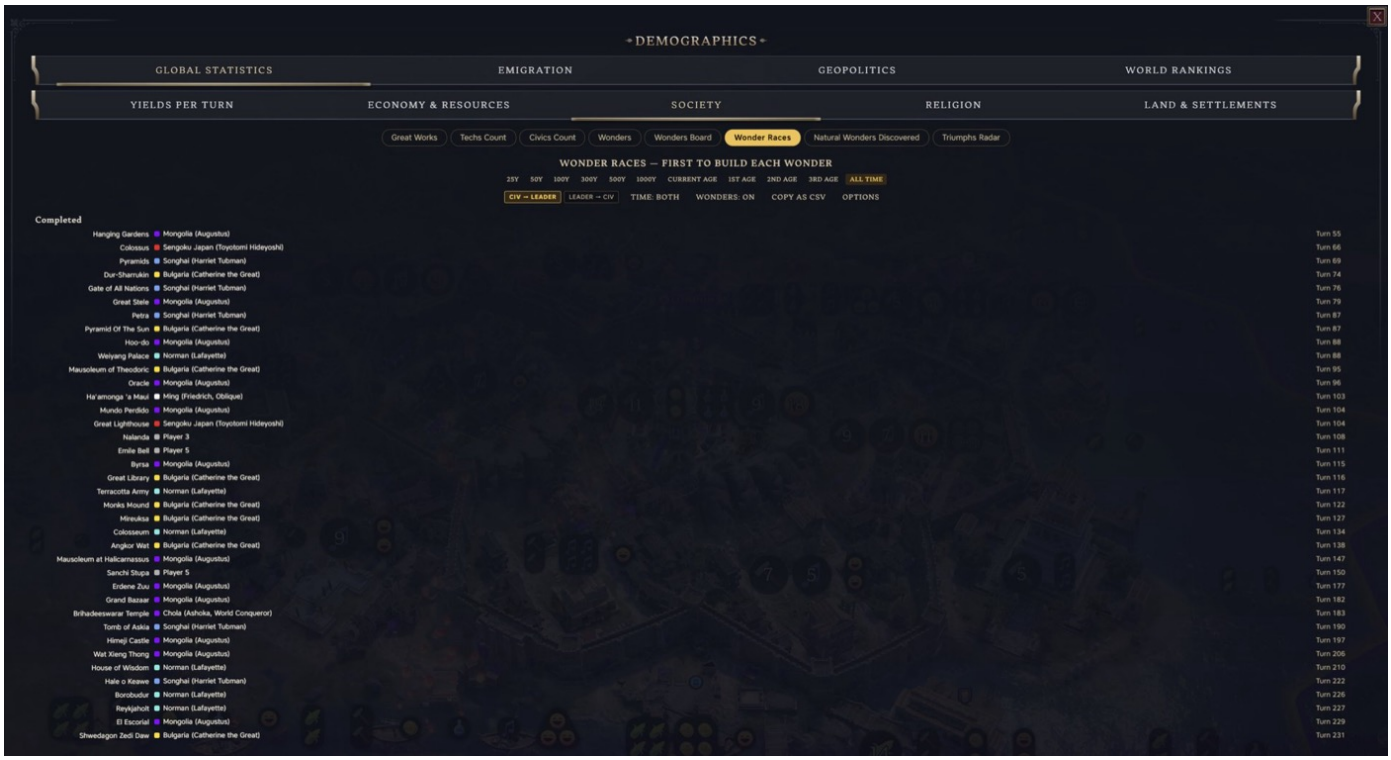

Figure 5: Wonder Races: who built each wonder first, and when

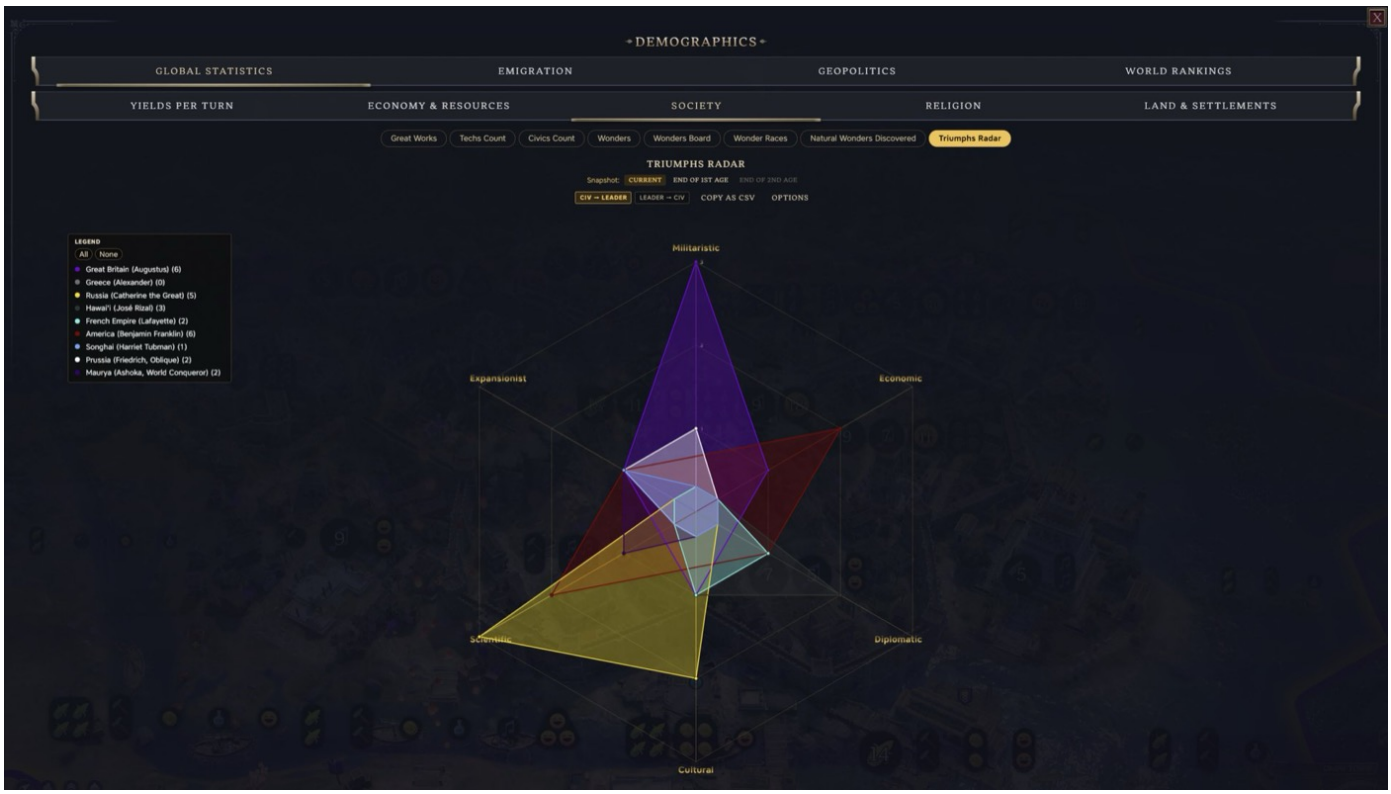

Figure 6: Triumphs radar: Legacy Path progress side by side

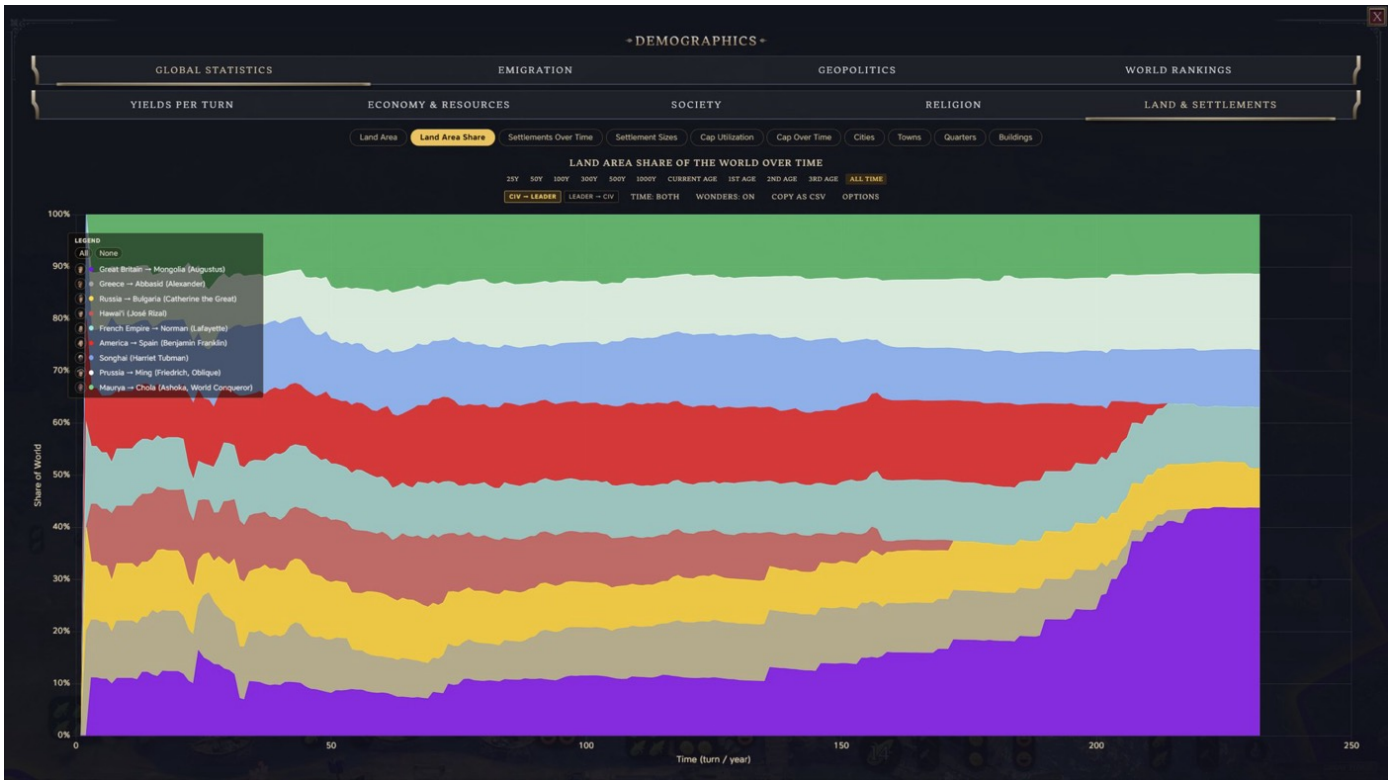

Figure 7: Share of the world's land over time

Figure 8: Settlement sizes and the most urbanized settlements

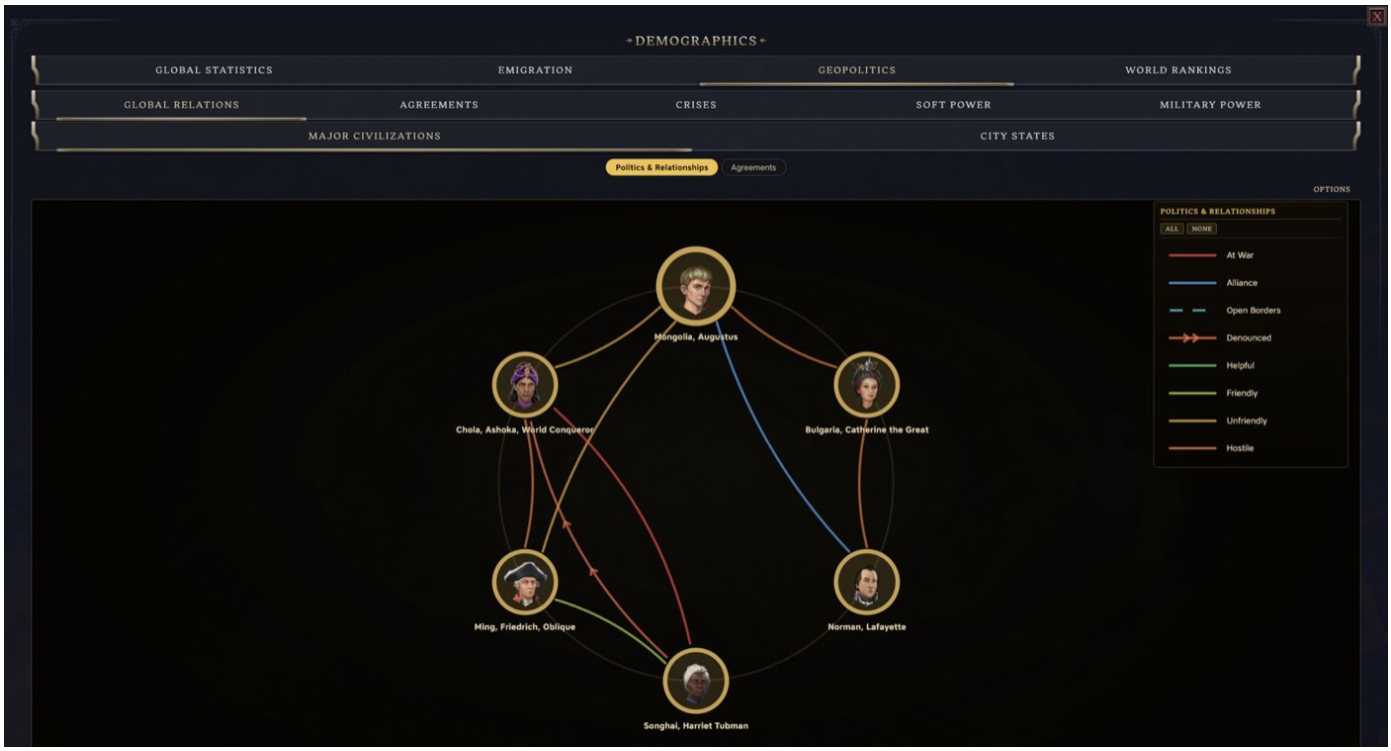

Figure 9: Global Relations: wars, alliances, and attitudes between met civilizations

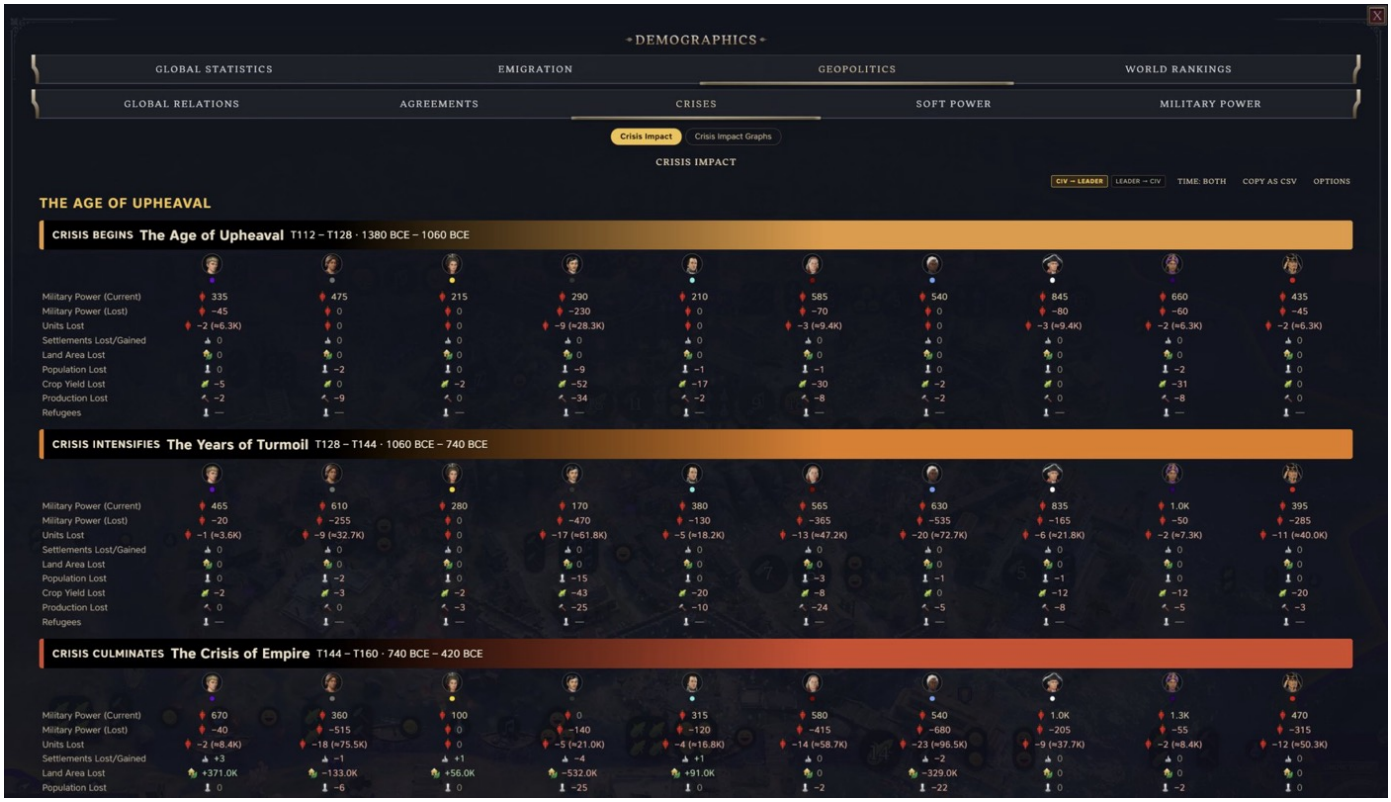

Figure 10: Crisis Impact: what each stage of the crisis cost every civilization

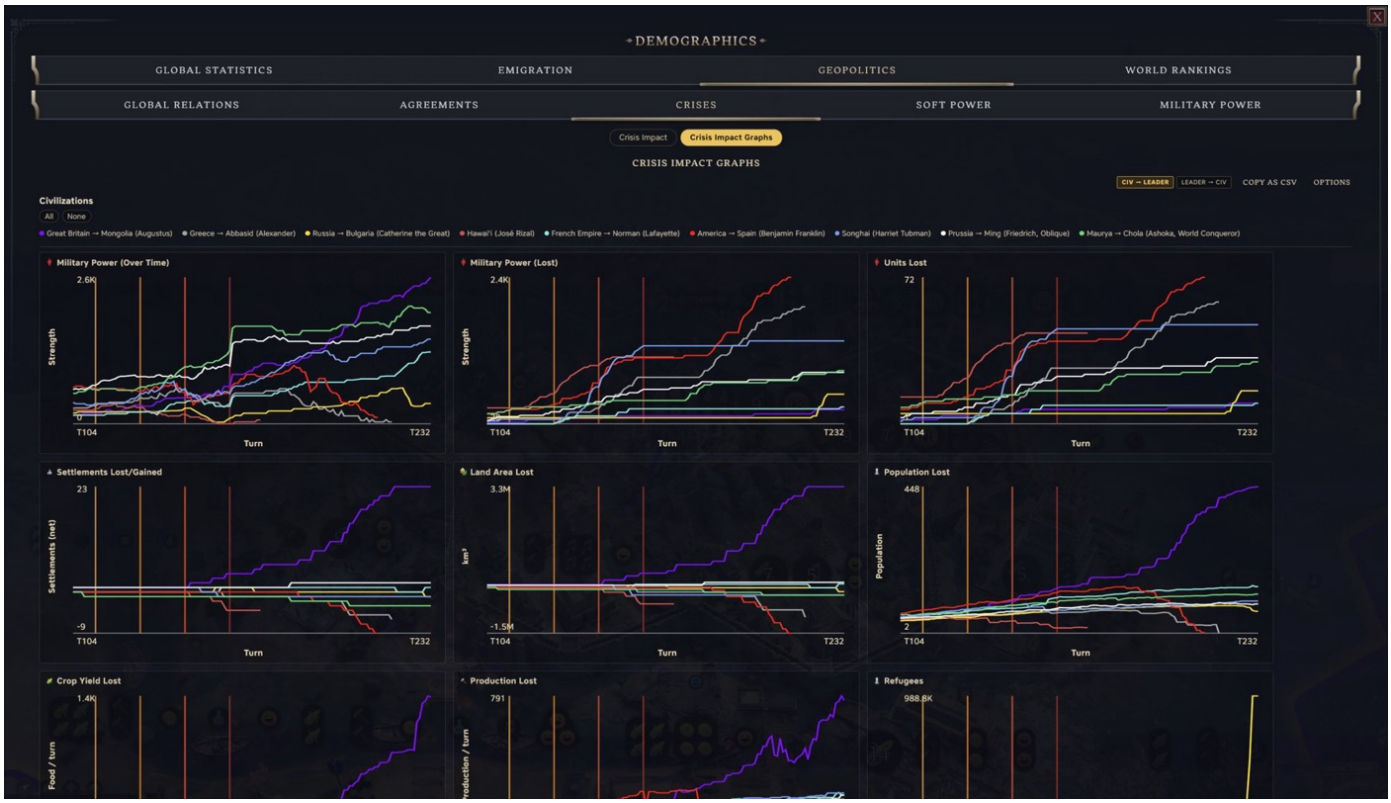

Figure 11: Crisis Impact Graphs: each loss charted across the crisis

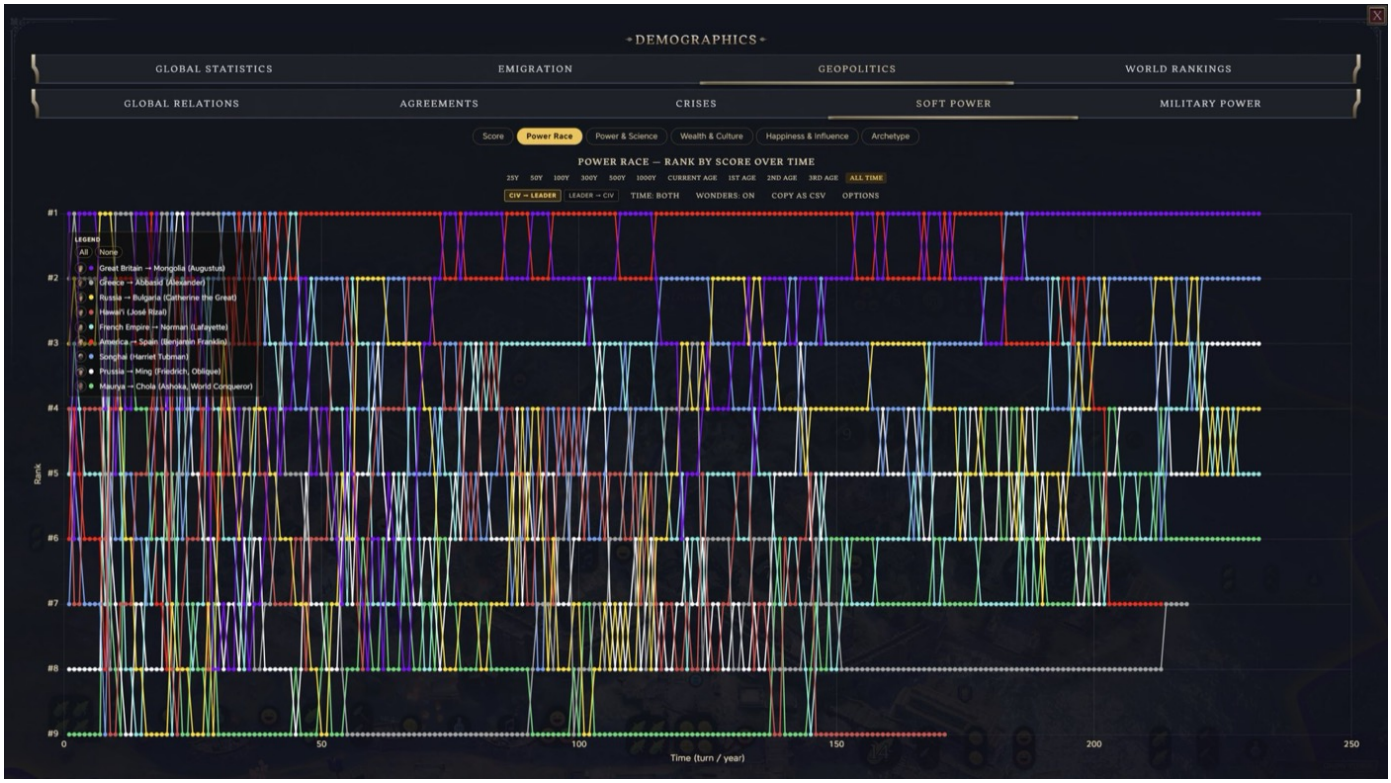

Figure 12: Power Race: every civilization's score rank, turn by turn

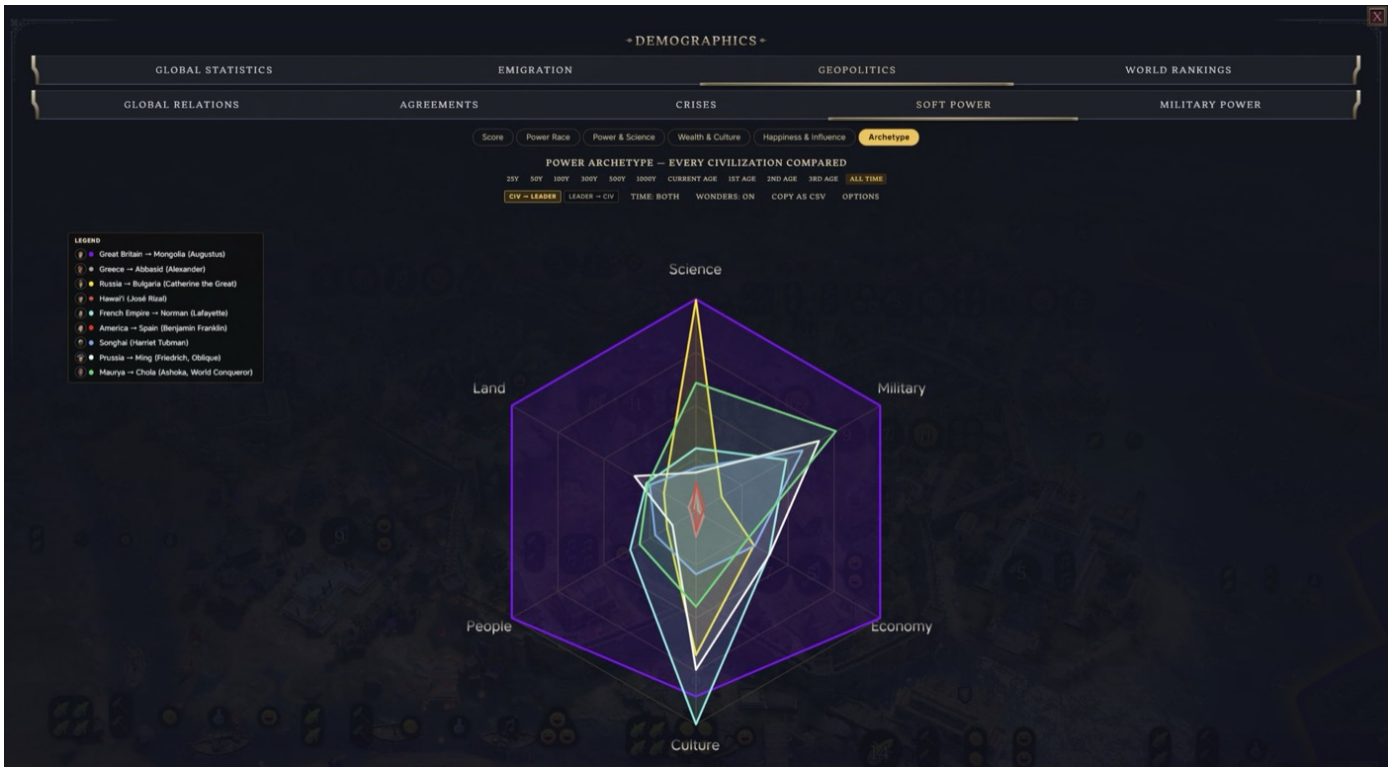

Figure 13: Archetype: science, military, economy, culture, people, and land compared

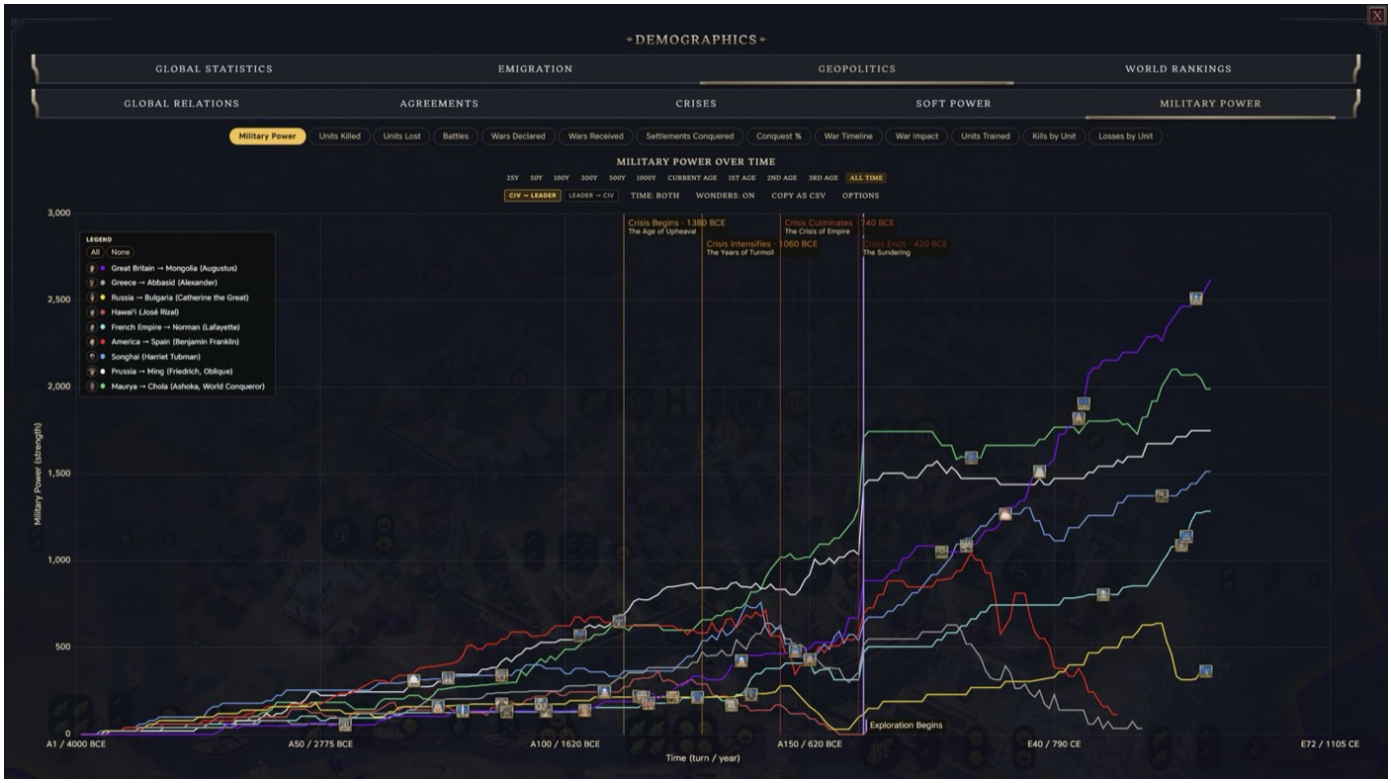

Figure 14: Military power over time

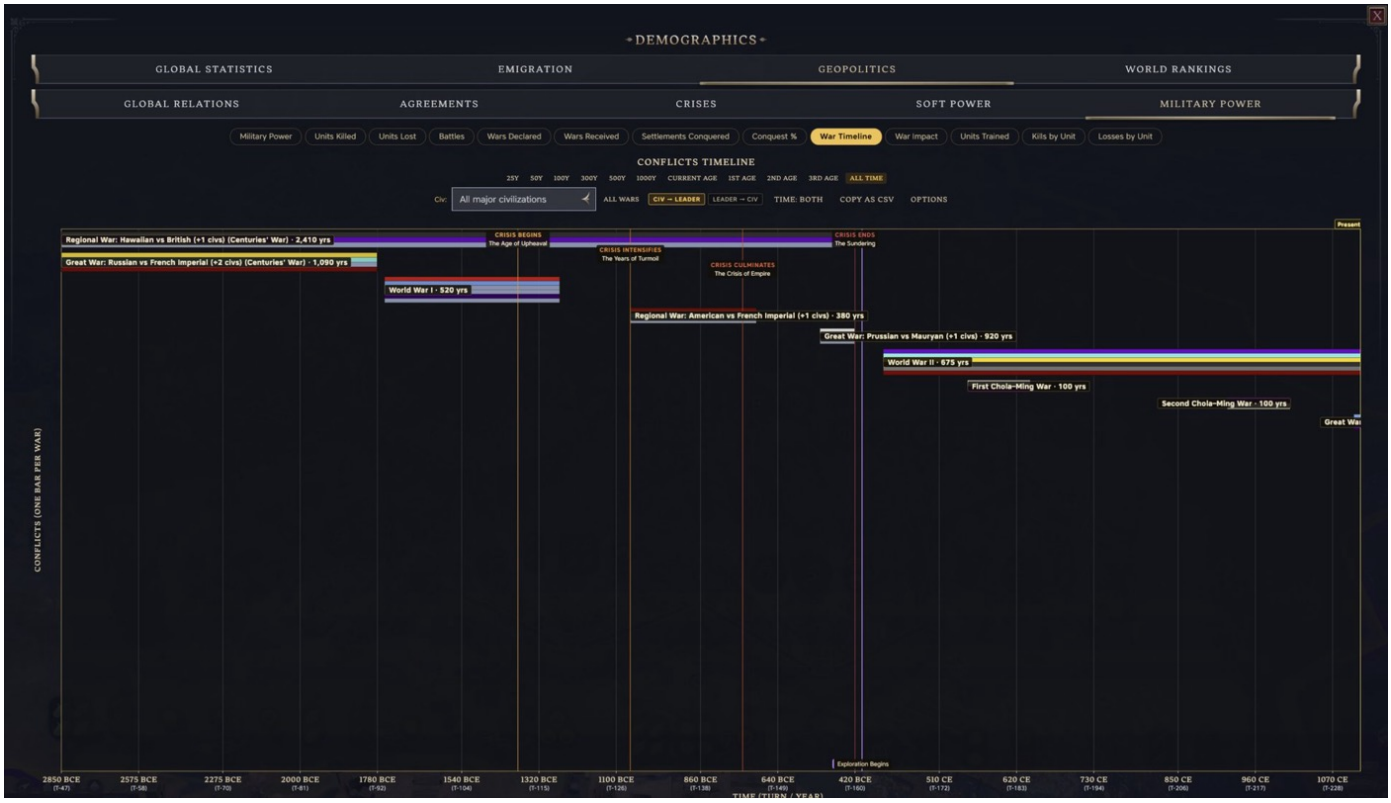

Figure 15: War Timeline: every war this game, with its generated name

Figure 16: Civilization Ranking

Figure 17: Civilization Rank by Yield: rates and yields, with Totals & Tallies below

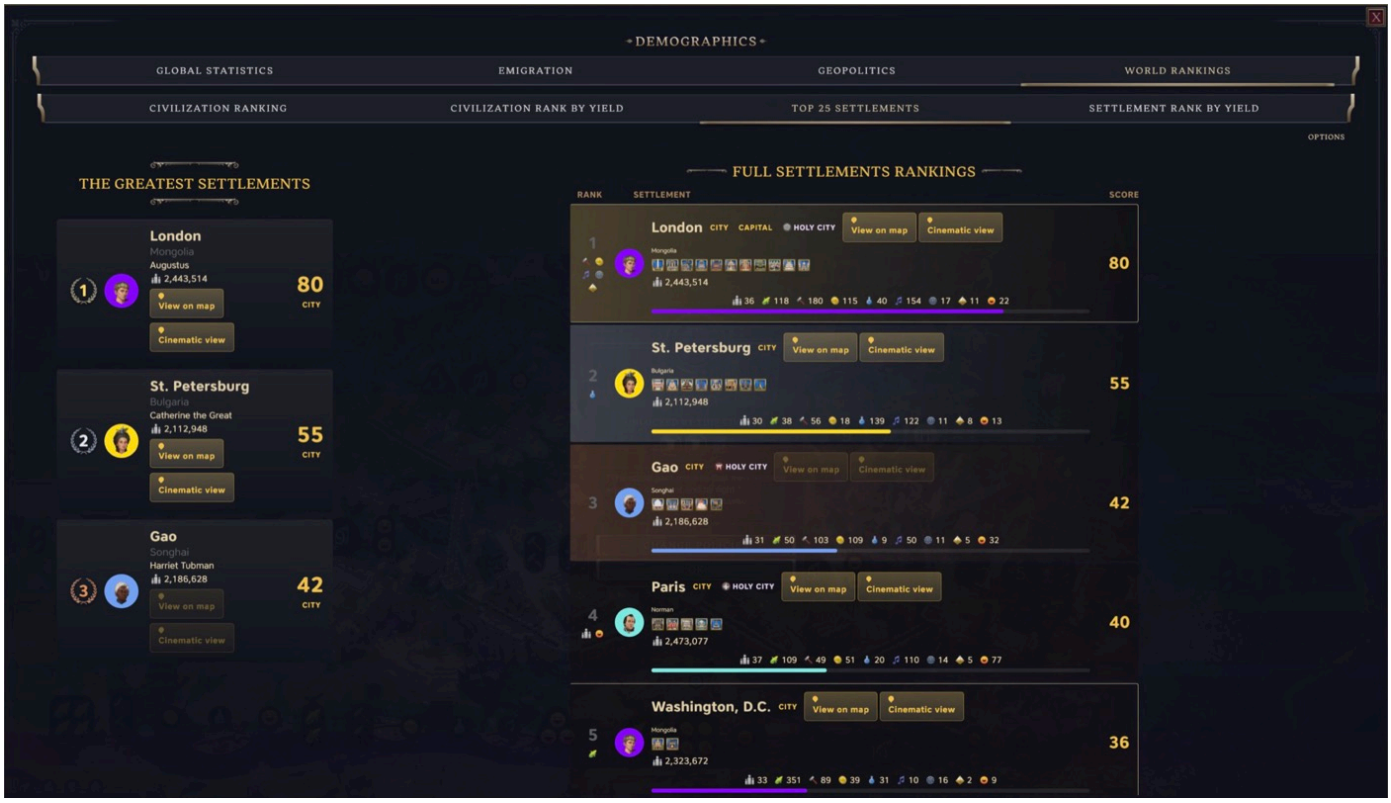

Figure 18: Top 25 Settlements: holy cities, world leaders, and map fly-to

Rank	Owner	Settlement	Type	Score	Population	Food	Production	Gold	Science	Culture	Influence	Wonders	Happiness
1	Mongolia Augustus	London	CITY	80	36	118	180	115	40	154	17	11	22
2	Bulgaria Catherine the Great	St. Petersburg	CITY	55	30	38	56	18	139	122	11	8	13
3	Songhai Harriet Tubman	Gao	CITY	42	31	50	103	109	9	50	11	5	32
4	Norman Lafayette	Paris	CITY	40	37	109	49	51	20	110	14	5	77
5	Mongolia Augustus	Washington, D.C.	CITY	36	33	351	89	39	31	10	16	2	9
6	Norman Lafayette	Rouen	CITY	28	29	69	57	41	14	127	11	1	77
7	Chola Ashoka, World Conqueror	Pataliputra	CITY	25	35	76	58	48	43	43	9	1	32
8	Ming Friedrich, Oblique	Yingtian	CITY	24	19	90	63	40	50	53	8	0	30
9	Ming Friedrich, Oblique	Berlin	CITY	23	23	84	50	47	35	39	6	1	13
10	Chola Ashoka, World Conqueror	Gangaikonda Cholapuram	CITY	22	35	74	79	29	61	34	10	0	23
11	Bulgaria Catherine the Great	Preslav	CITY	21	16	27	21	26	76	73	8	0	30
12	Unmet Civilization Unmet Leader	Unknown Settlement	CITY	20	25	74	61	32	9	22	4	2	30
13	Mongolia Augustus	Lāhainā	CITY	18	29	95	56	22	16	7	8	2	-9
14	Mongolia Augustus	Athēnai	TOWN	18	27	32	48	86	14	8	2	0	19
15	Songhai Harriet Tubman	Kabara	CITY	15	24	60	50	48	7	19	6	0	36

Figure 19: Settlement Rank by Yield, filterable to cities or towns

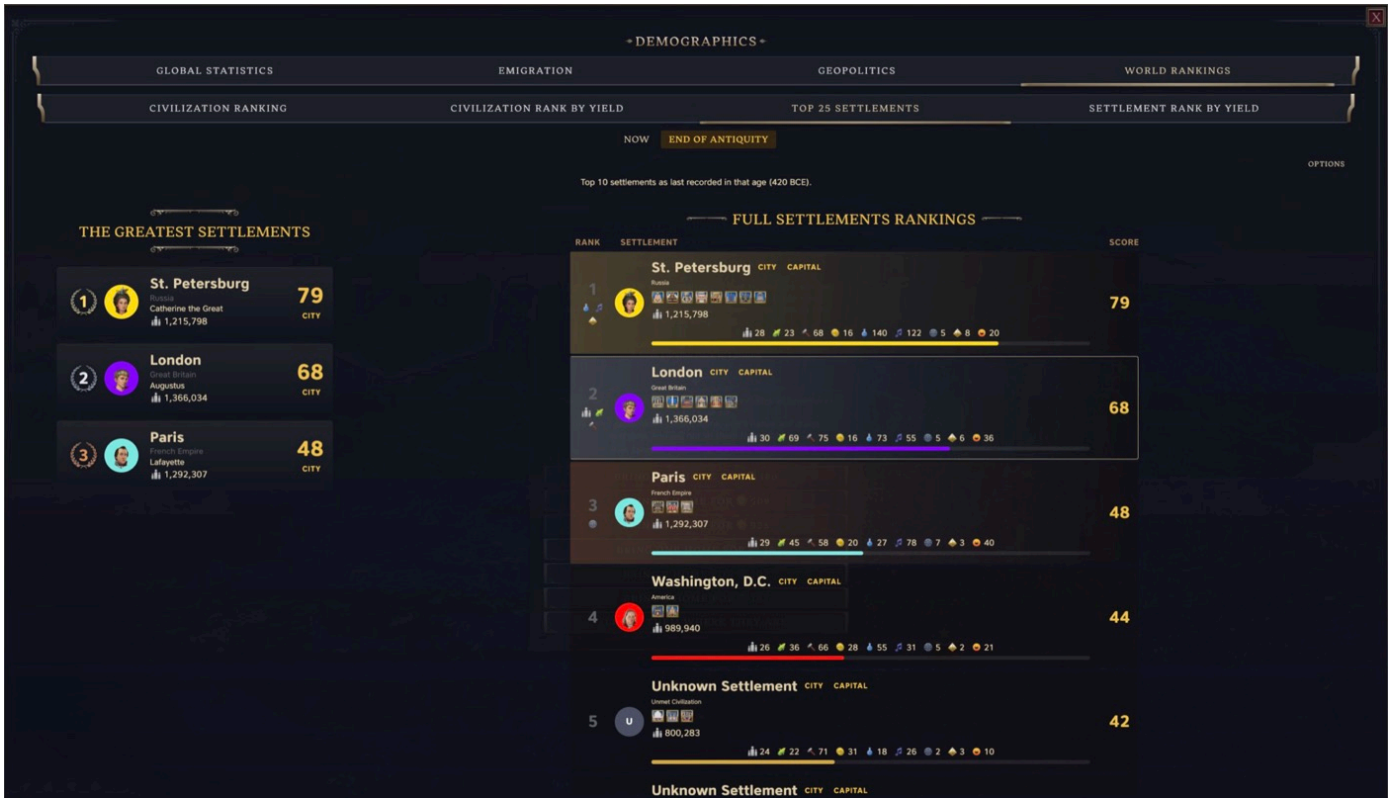

Figure 20: End of Antiquity: the age's final top ten, kept after the age ends

Figure 21: Totals & Tallies, the second Civilization Rank by Yield table

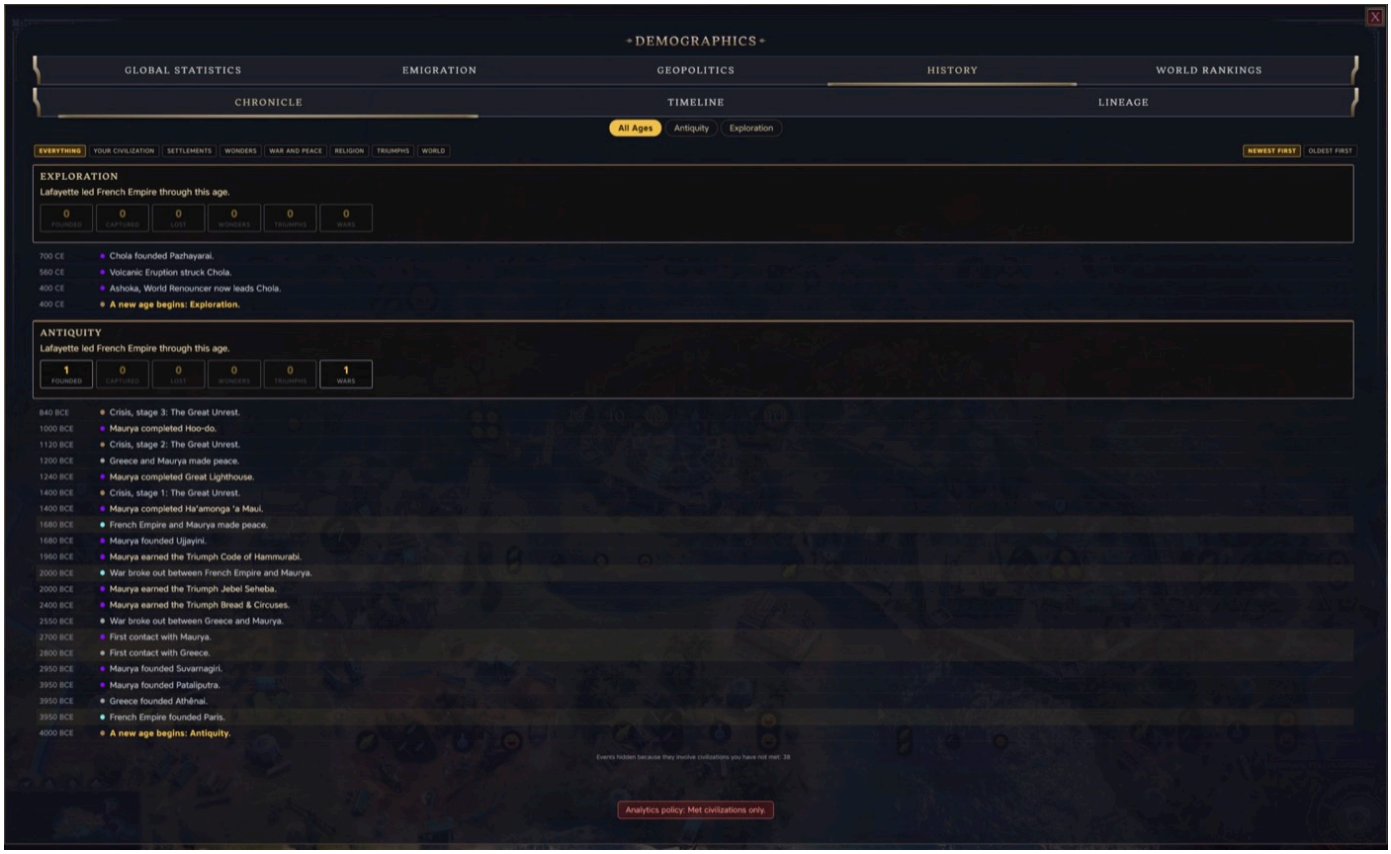

Figure 22: Chronicle: your campaign told age by age

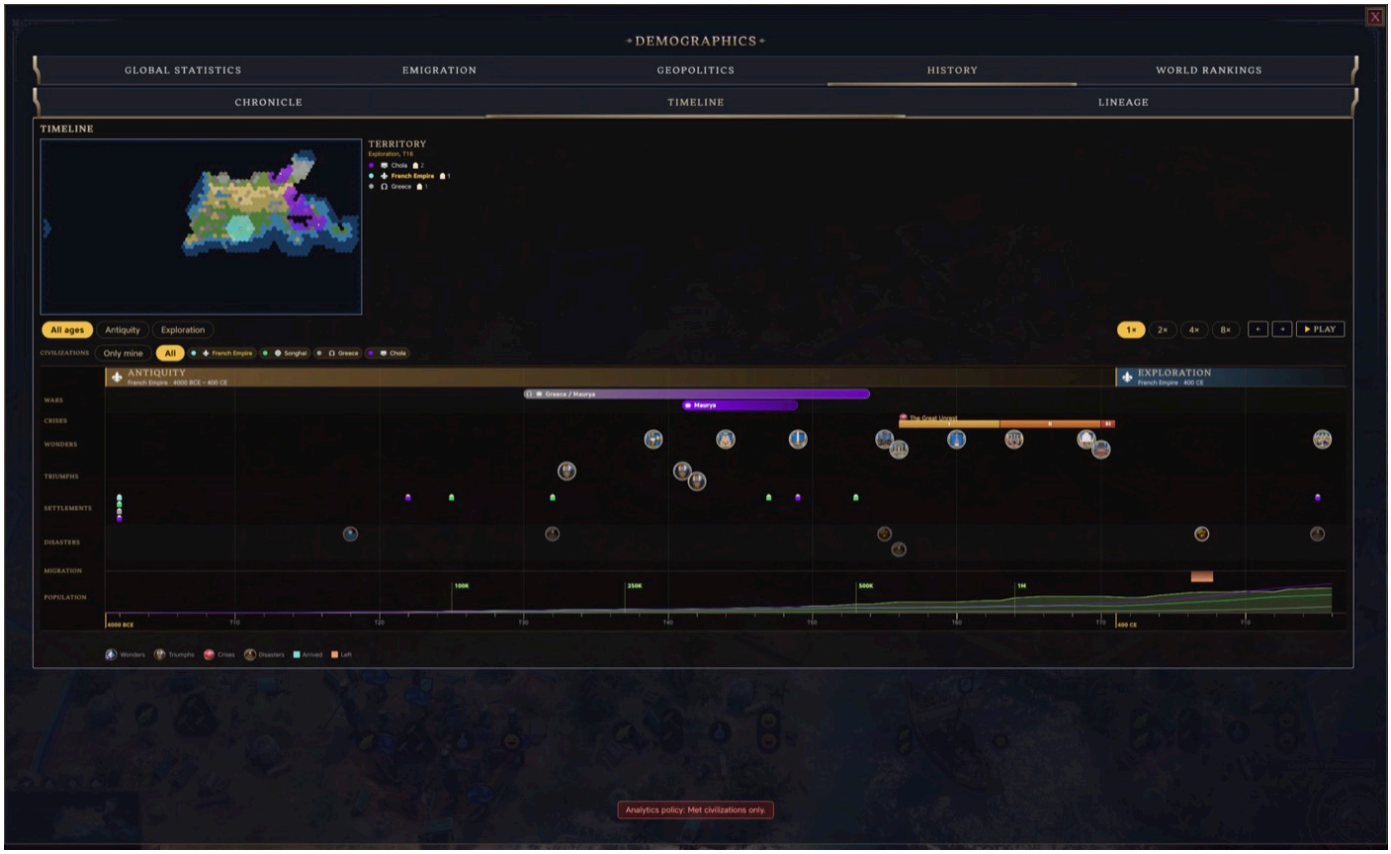

Figure 23: Timeline: wars, crises, wonders, Triumphs, settlements, disasters and population, with the territory map

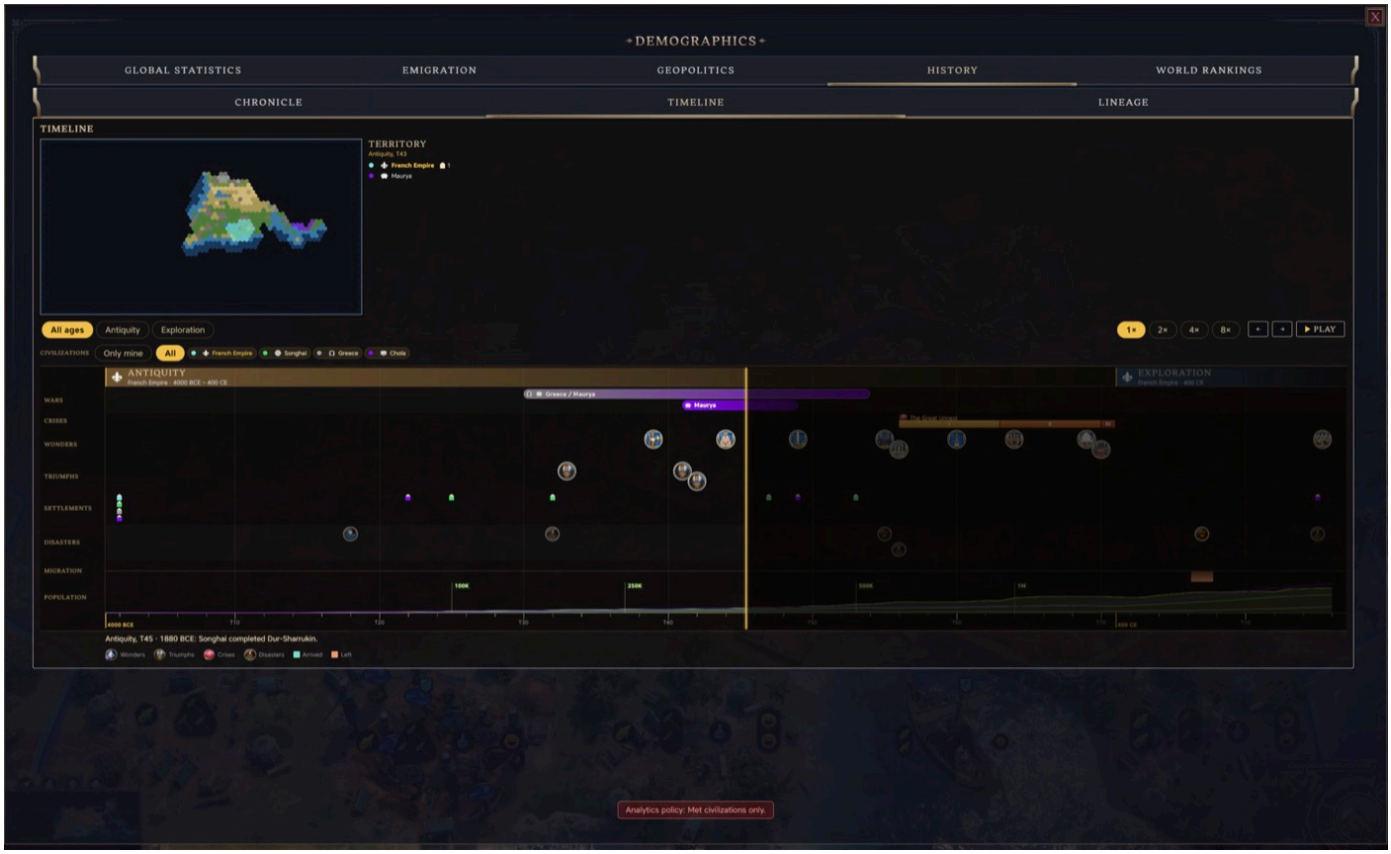

Figure 24: Play replays the game: the map and caption follow the cursor

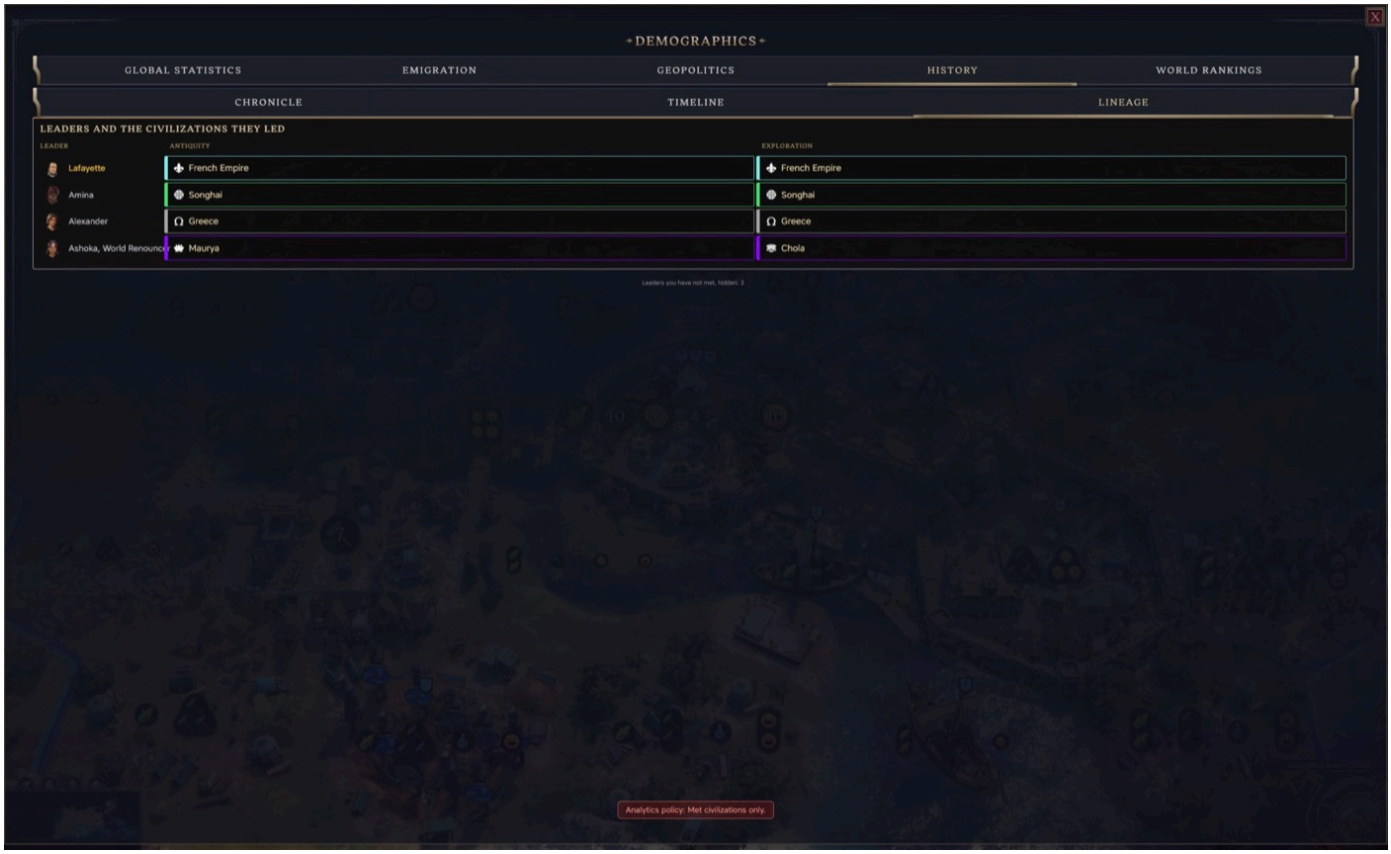

Figure 25: Lineage: every leader and the civilizations they led

Figure 26: Best Games: your finest playthroughs, and this game in context

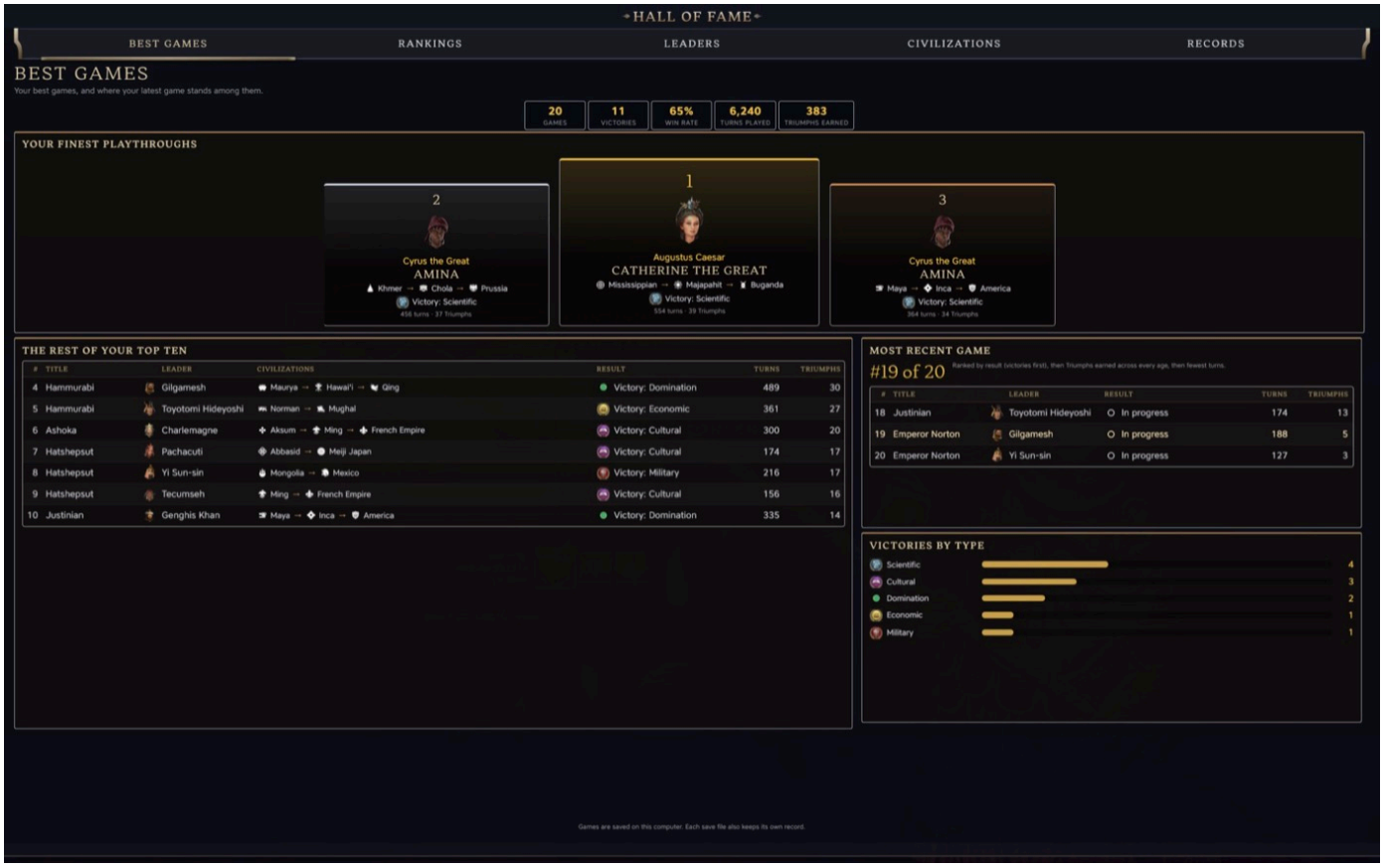

Figure 27: Best Games from the main menu, with your most recent game

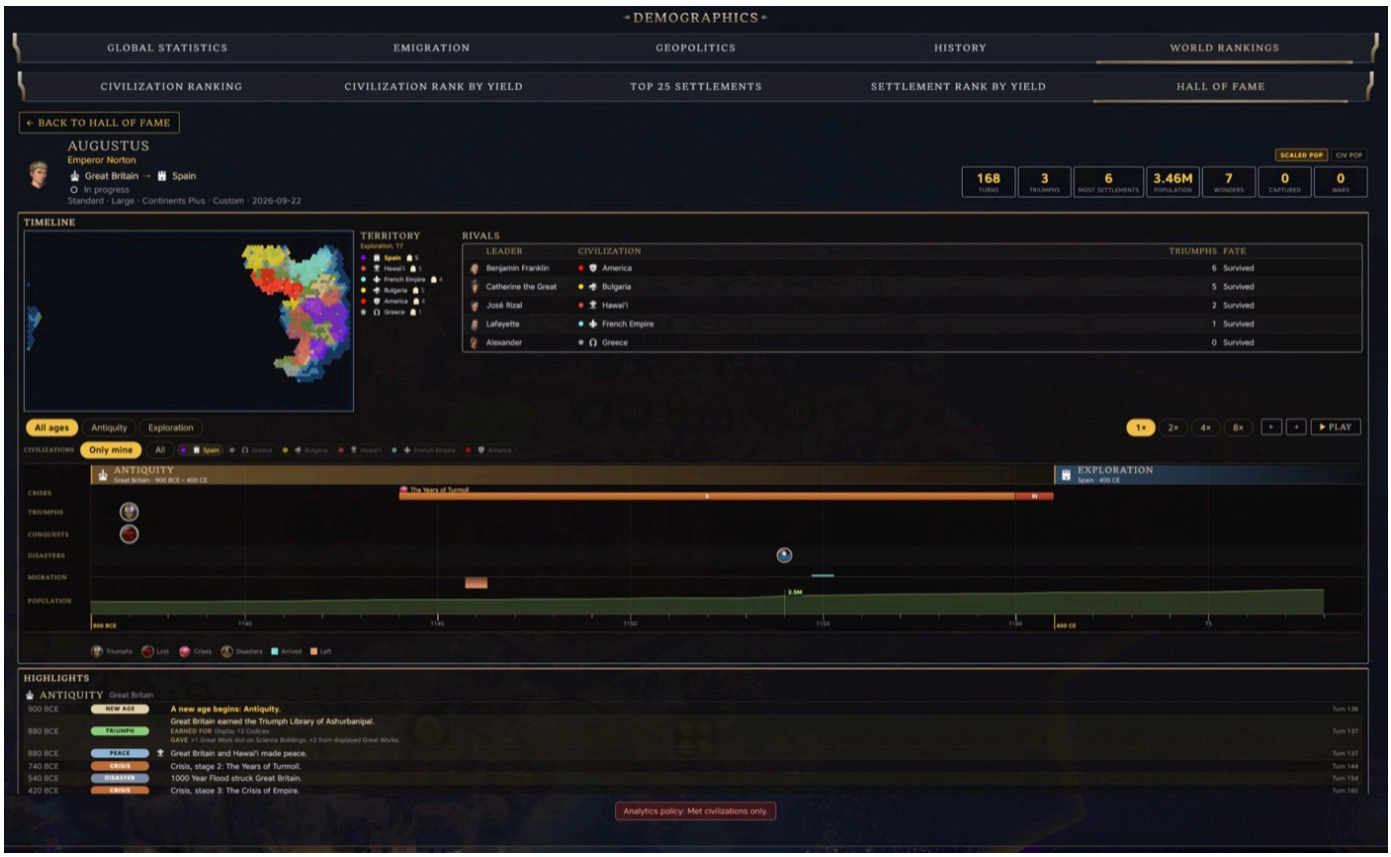

Figure 28: A game's page in the Hall of Fame

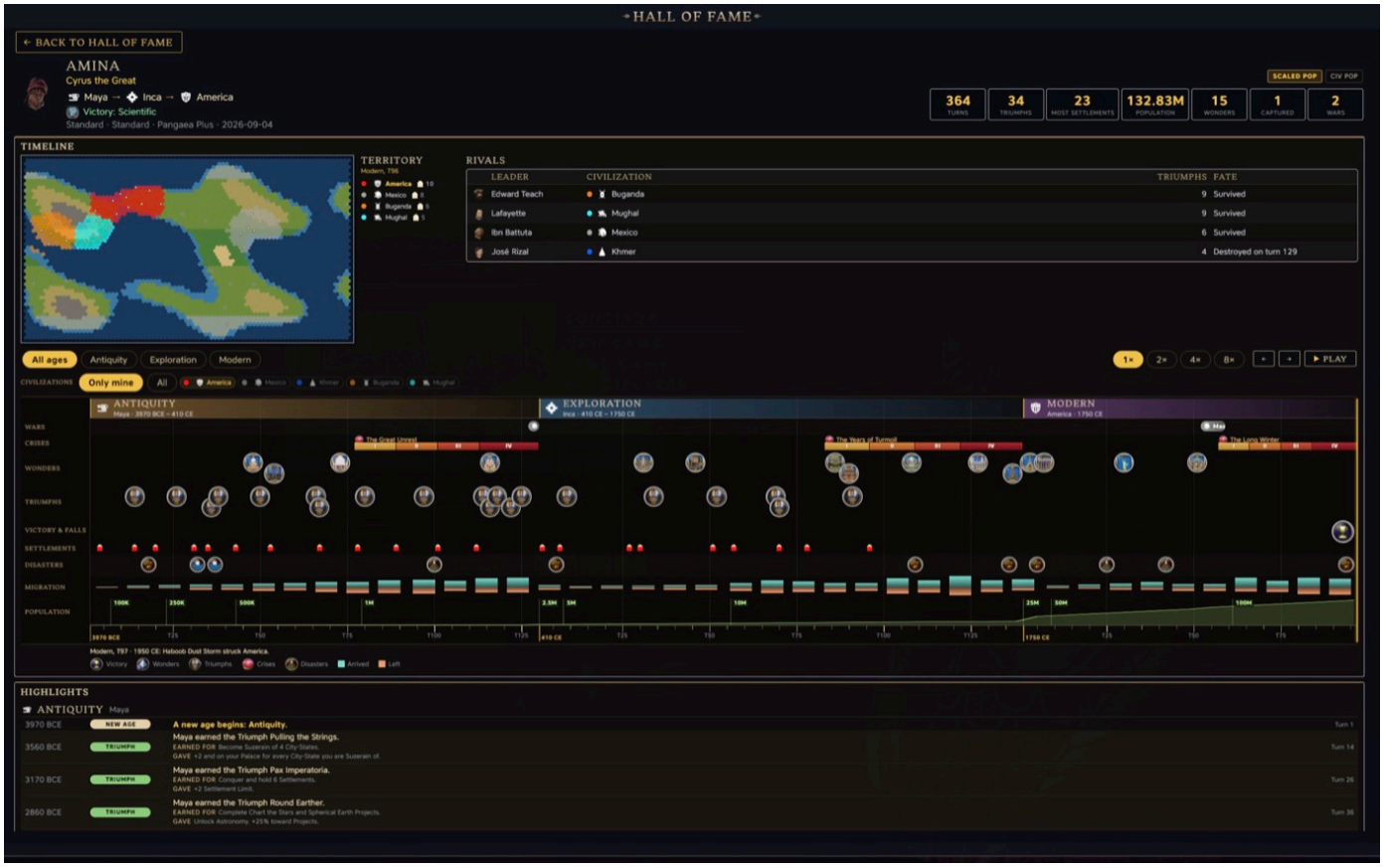

Figure 29: The same page from the main menu, no game loaded

9. Companion-mod integration
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1. Layout

The dashboard has three rows of navigation:

- **Top tabs** pick a hub: **Global Statistics**, **Geopolitics**, **History**, and **World Rankings**. When the *Emigration* companion mod is installed, an **Emigration** tab appears beside them.
- **Page tabs** pick a page within the hub (for example *Yields Per Turn* or *Military Power*).
- **Metric pills** pick the chart on that page.

Every chart page shares one toolbar: a time-range filter (25 to 1000 years, current age, a single age, or all time), a **Civ → Leader / Leader → Civ** label order, a time-axis mode, wonder markers on or off, **Copy as CSV**, and an **Options** button. Charts mark each crisis stage and age change as a vertical line, and wonders as icons on the line of the civilization that built them. The legend toggles civilizations on and off.

Settings live in the game's own **Options** screen, under **Mods → Demographics**. The **Options** button opens it directly on top of the dashboard.

2. Global Statistics

Per-turn history for every met civilization, on five pages:

- **Yields Per Turn:** Gold, Production, Food, Science, Culture, Influence, and Happiness per turn.

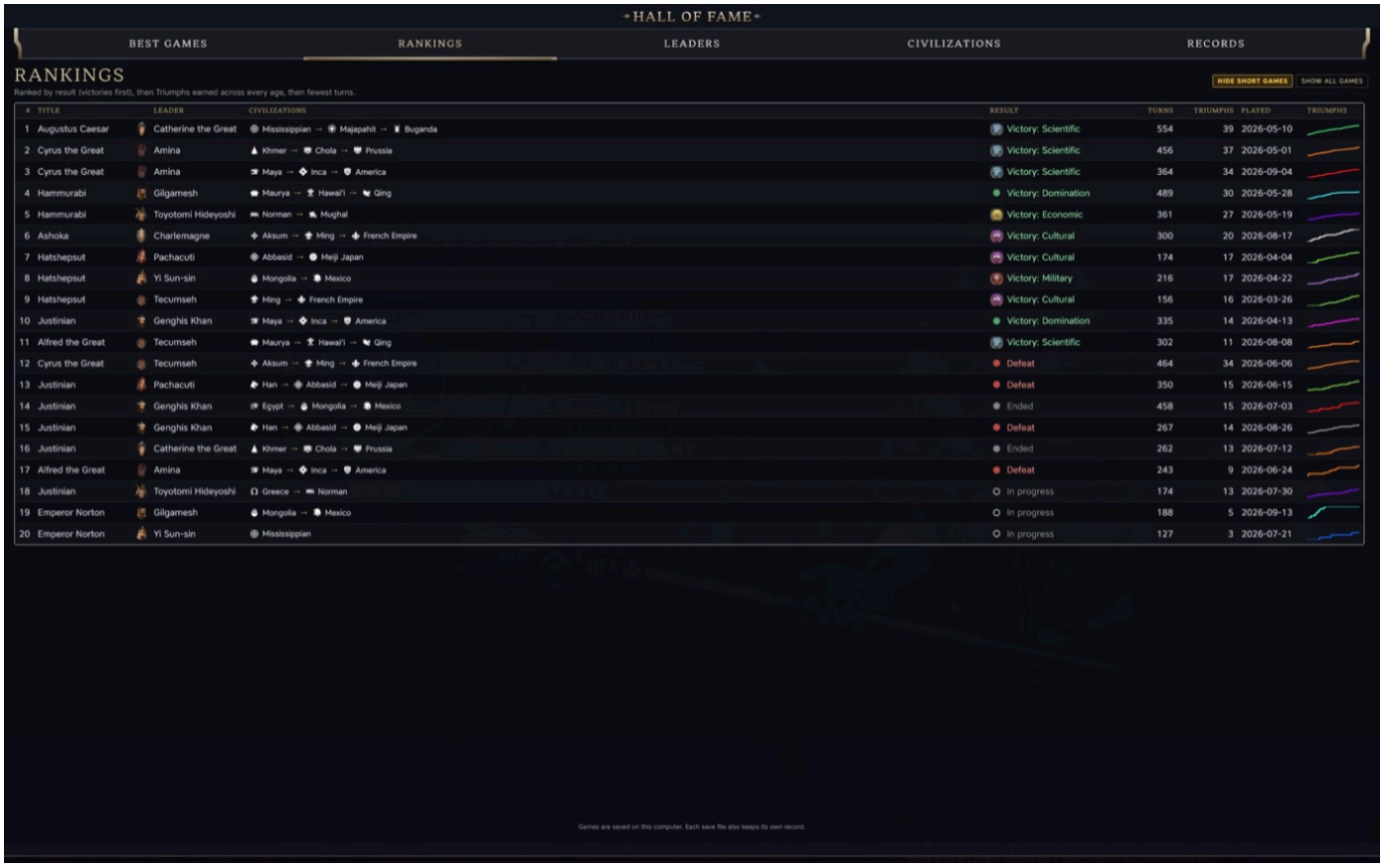

Figure 30: Rankings, each game with its honorific

- **Economy & Resources:** GDP, Treasury, Trade Routes, Total Resources, a stacked chart of one civilization's resources by type, and separate Bonus, Empire, and City resource counts, plus Treasure resources in Exploration and Factory resources in the Modern age.
- **Society:** Great Works, Techs, Civics, Wonders, a **Wonders Board** listing each civilization's wonders with the turn they were built, **Wonder Races** showing who built each wonder first, Natural Wonders Discovered, and a **Triumphs Radar** of Legacy Path progress. Faith, Tourism, and Great People appear once any civilization has recorded a value for them.
- **Religion:** in Antiquity, the pantheons each civilization chose and their yields; from Exploration on, religion standings (settlements following each religion), religious spread over time, and followers by population.
- **Land & Settlements:** Land Area, each civilization's share of the world's land, settlements over time, a settlement-size distribution with the most urbanized settlements, settlement-cap use and the cap over time, city and town counts, and boards of quarters and buildings constructed by type.

Without *Emigration*, **Population** sits on the Society page. With it, Population and Population Share move to the Emigration tab's first page, beside the migration charts.

3. Geopolitics

- **Global Relations:** a ring with your civilization and the civilizations you have met, switchable to **City States**. Edges show wars, alliances, open borders, denouncements, and attitudes; a second view, **Agreements**, draws the active agreements between civilizations. Colorblind mode switches to a high-contrast palette.
- **Agreements:** Diplomatic Approval and Ongoing Deals over time.
- **Crises: Crisis Impact** (per-stage cost tables, see §6) and **Crisis Impact Graphs** (each crisis statistic charted per civilization).
- **Soft Power:** Score, the **Power Race** (every civilization's score rank, turn by turn), three scatter plots (Power & Science, Wealth & Culture, Happiness & Influence), and an **Archetype** radar comparing science, military, economy, culture, people, and land.

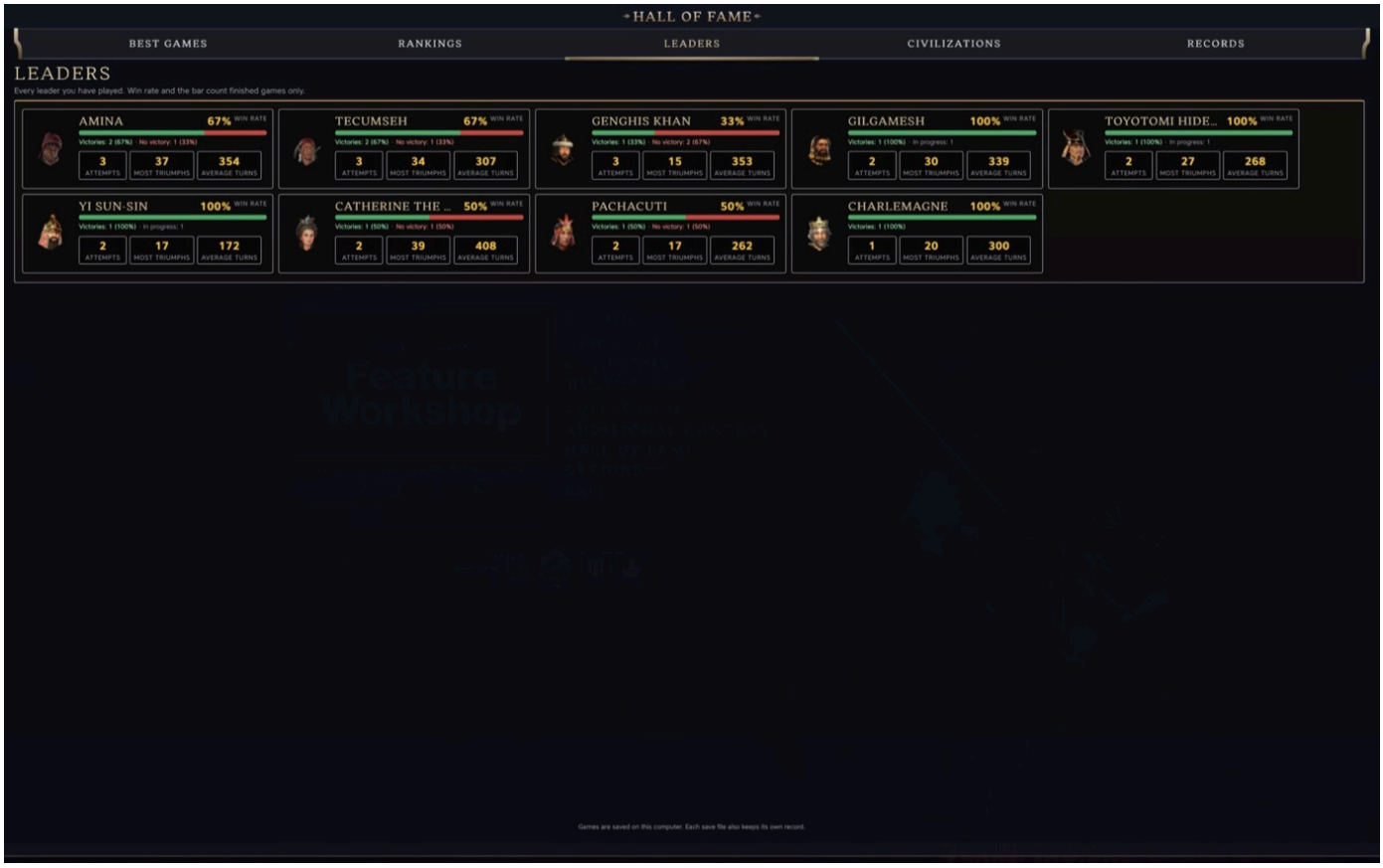

Figure 31: Leaders you have played

- **Military Power:** Military Power, Units Killed, Units Lost, Battles, Wars Declared, Wars Received, Settlements Conquered, Conquest %, the **War Timeline**, **War Impact** (see §6), and boards of units trained, kills, and losses by unit type.

4. History

Three pages, recorded from the moment Demographics is active in a game (the Timeline page also carries the civilizations filter and the territory map):

- **Chronicle:** the campaign's history, grouped by age. Each age opens with a summary of your civilization's deeds (settlements founded, captured and lost, wonders, Triumphs, wars, the religion you founded, and civilizations destroyed), followed by every recorded event in order: settlements founded, taken or lost, wonders completed, wars and peace, religions founded, Triumphs earned, crisis stages, natural disasters that struck a civilization's land, first contacts, eliminations, new ages and victories. Pills filter by age and by kind of event, and the order can run newest or oldest first. Events involving civilizations you have not met are hidden and counted, following the analytics policy.
- **Timeline:** the whole game on one scrolling chart. A banner for each age carries your civilization's emblem and the age's dates. Below it run lanes for your wars (in the enemy's color, with their emblem) and the age crises stage by stage, then one lane per kind of milestone (**Wonders**, **Triumphs**, **Religion**, **Conquests**, **Victory & falls**) drawn with the game's own icons, such as each wonder's art and each religion's symbol, then every settlement founded, natural disasters, migration in and out (with the *Emigration* mod), the population curve with its milestones, and a turn ruler. Pills show all ages or one, zoom runs from 1× to 8× with pan buttons, clicking the ruler moves the cursor, and **Play** sweeps through the game, naming each event as it passes. Everything on the chart has a tooltip, and hovering the population lane or the ruler reads out the turn and population under the mouse.
- **Civilizations filter:** the row above the timeline picks whose story it shows. Every game opens on your own civilization; add any civilization you met, or **All**, to put their milestones, settlements, wars with each other,

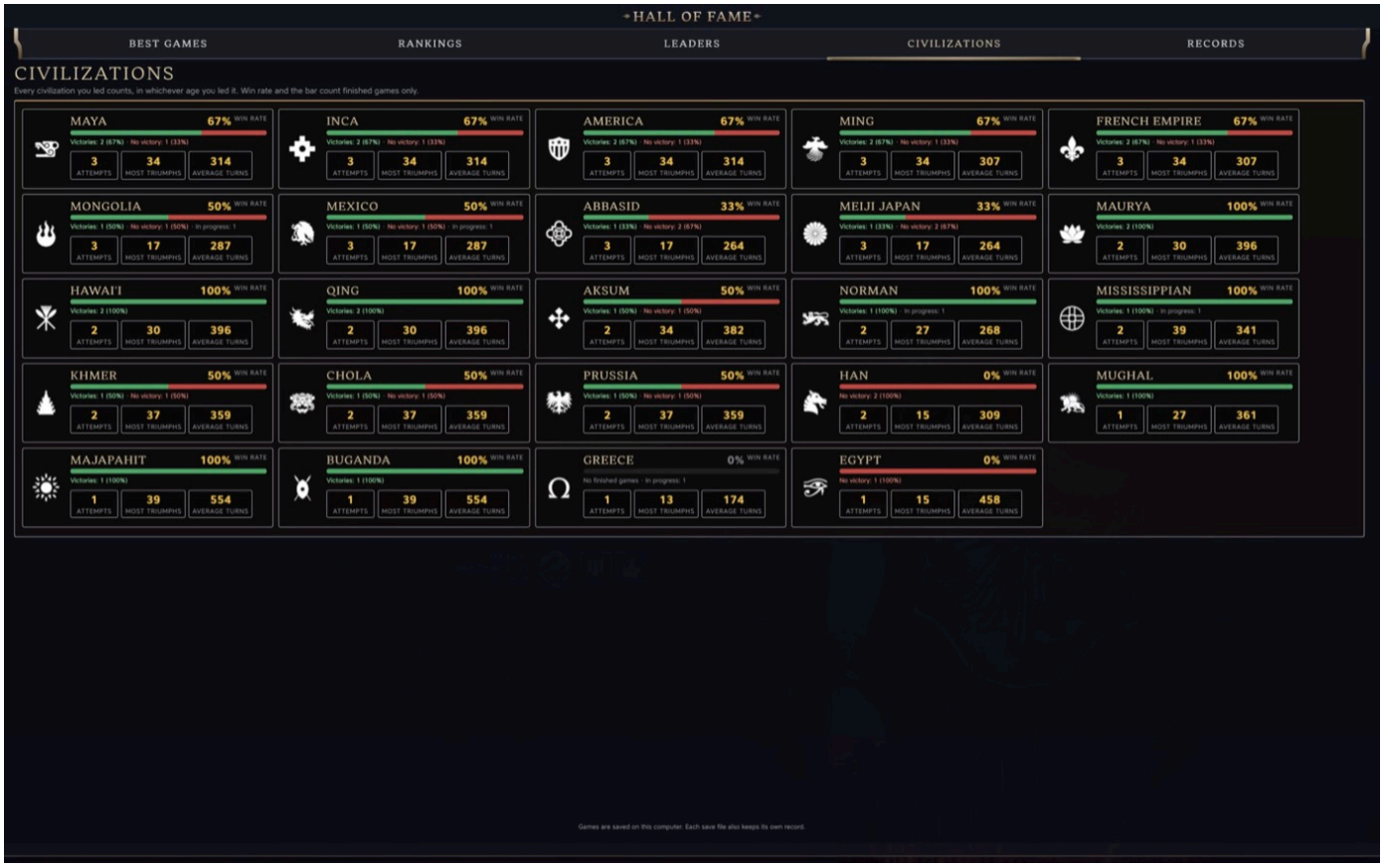

Figure 32: Civilizations you have played

disasters on their land and population line on the same lanes.

- **Territory map:** above the timeline, a hex map of the world laid out like the game's own minimap, one cell per tile: terrain by biome, each civilization's land tinted in its color, independent land in grey, settlements marked, and land you had not yet explored left blank. It follows the timeline's cursor, so playback replays both the expansion and your exploration. Civilizations you have not met stay off the map.
- **Lineage:** one row per leader and one column per age, each cell the civilization that leader led in that age in the leader's color, with the turn a line ended.

When Demographics is added to a game already in progress, the chronicle starts from that turn and the first age's summary says so.

5. World Rankings

Five pages:

- **Civilization Ranking:** a podium of the three strongest civilizations, each with its capital and **View on map** / **Cinematic view** buttons for it, and the full list ordered by cumulative settlement score, with population and settlement counts.
- **Civilization Rank by Yield:** every civilization's rank (or value, via **Rank** / **Value**) in each tracked figure, in two tables that share one sort: the rates and yields first, then **Totals & Tallies** (land, cities and towns, conquests, units, great works, migration). Click any column to sort both.
- **Top 25 Settlements:** a podium and ranked list of the strongest settlements. Each row shows the settlement's wonders (hover one for the year it was completed), a **Holy City** badge with the religion's icon when it is a religion's holy city, and small icons under the rank for every output it leads the world in. **View on map** snaps the camera to the city and **Cinematic view** plays a smooth orbit of it; with the experimental flyby enabled in Options, a short keyframed flyby is also available. Once an age ends, a pill for it (for example **End of Antiquity**) shows that age's final top ten.
- **Settlement Rank by Yield:** a sortable table of every settlement's population and yields, filterable to **All**,

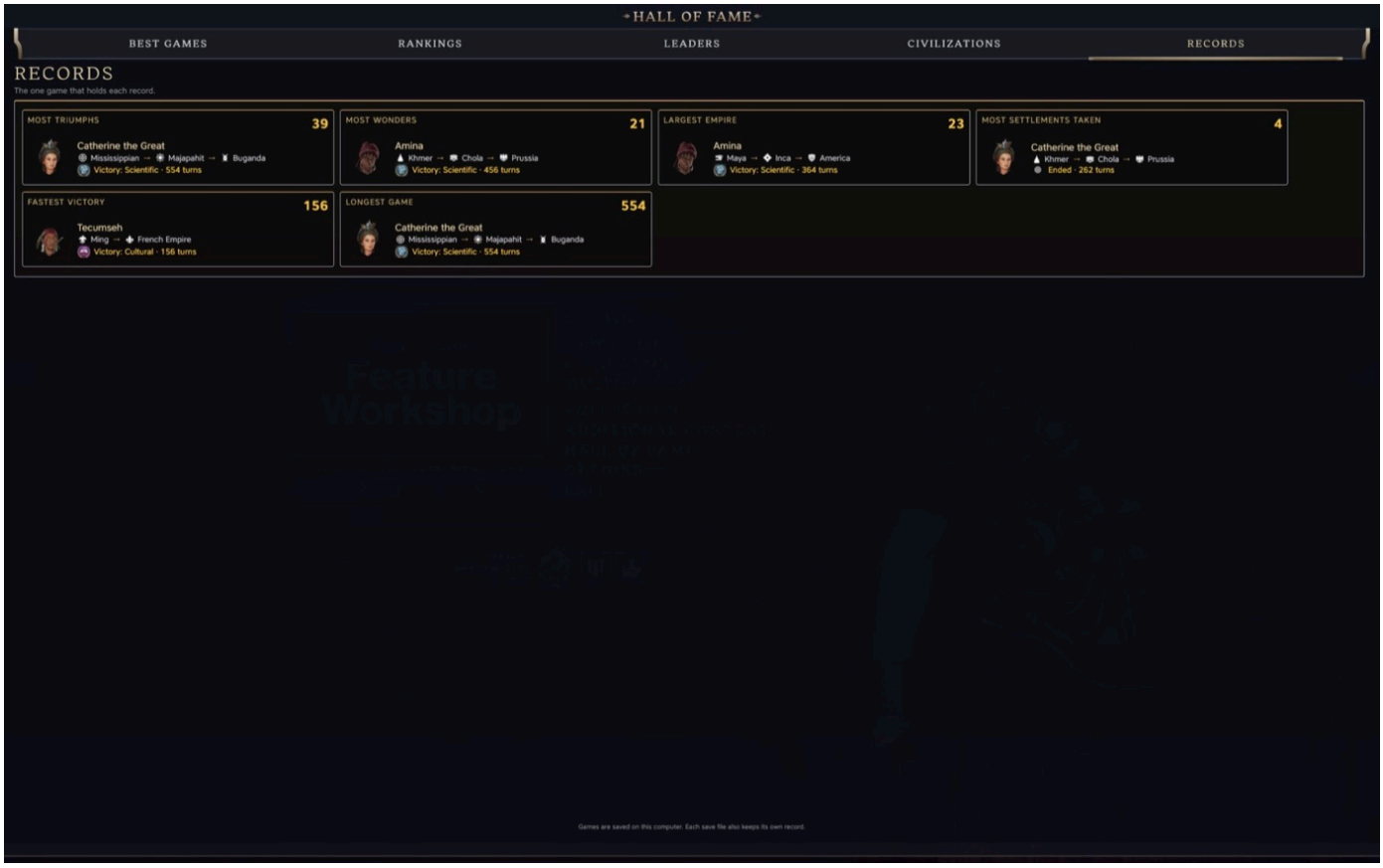

Figure 33: Records: the game that holds each one

Figure 34: Hall of Fame on the main menu

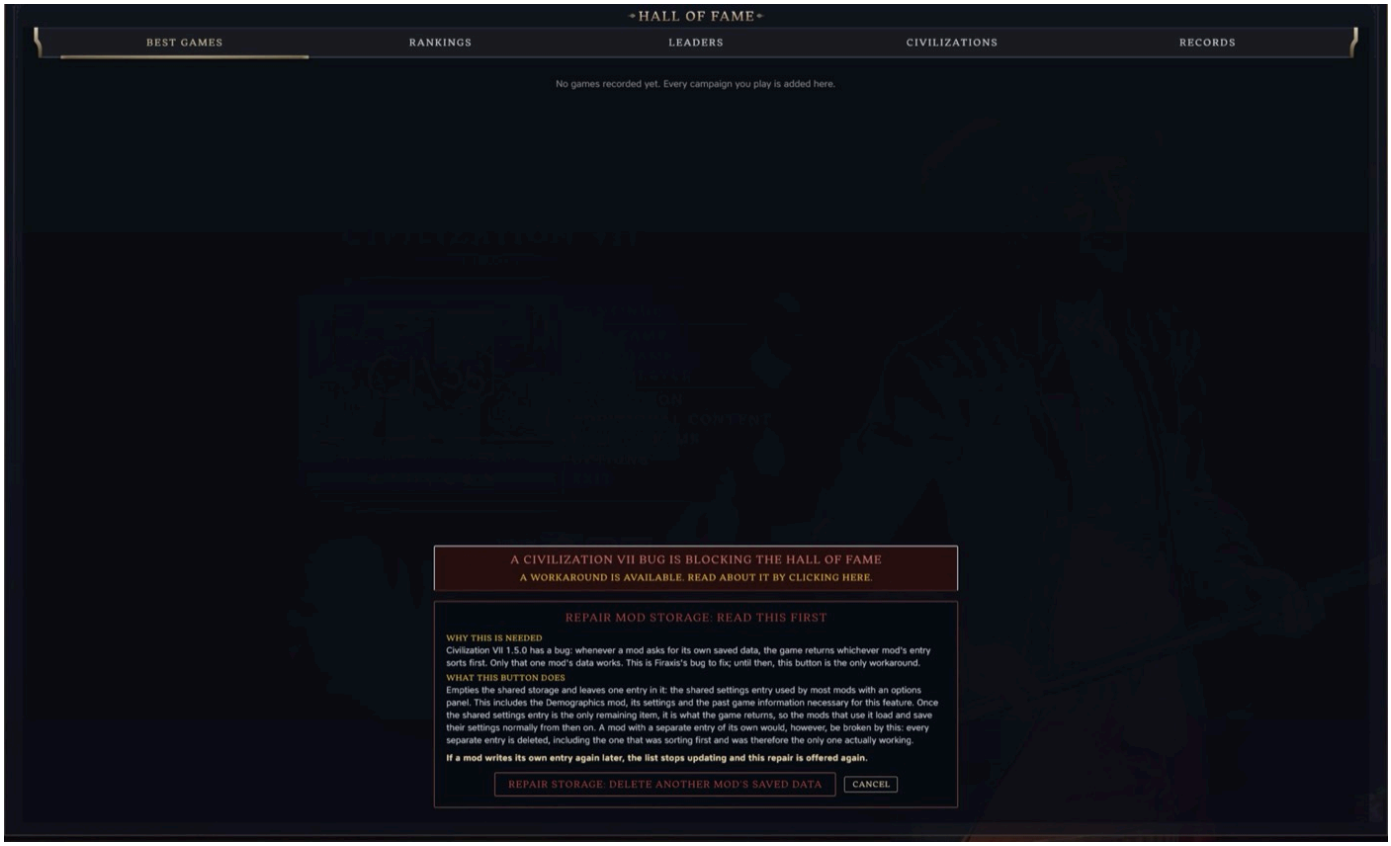

Figure 35: The notice when the game's storage bug blocks the list; the box is the button

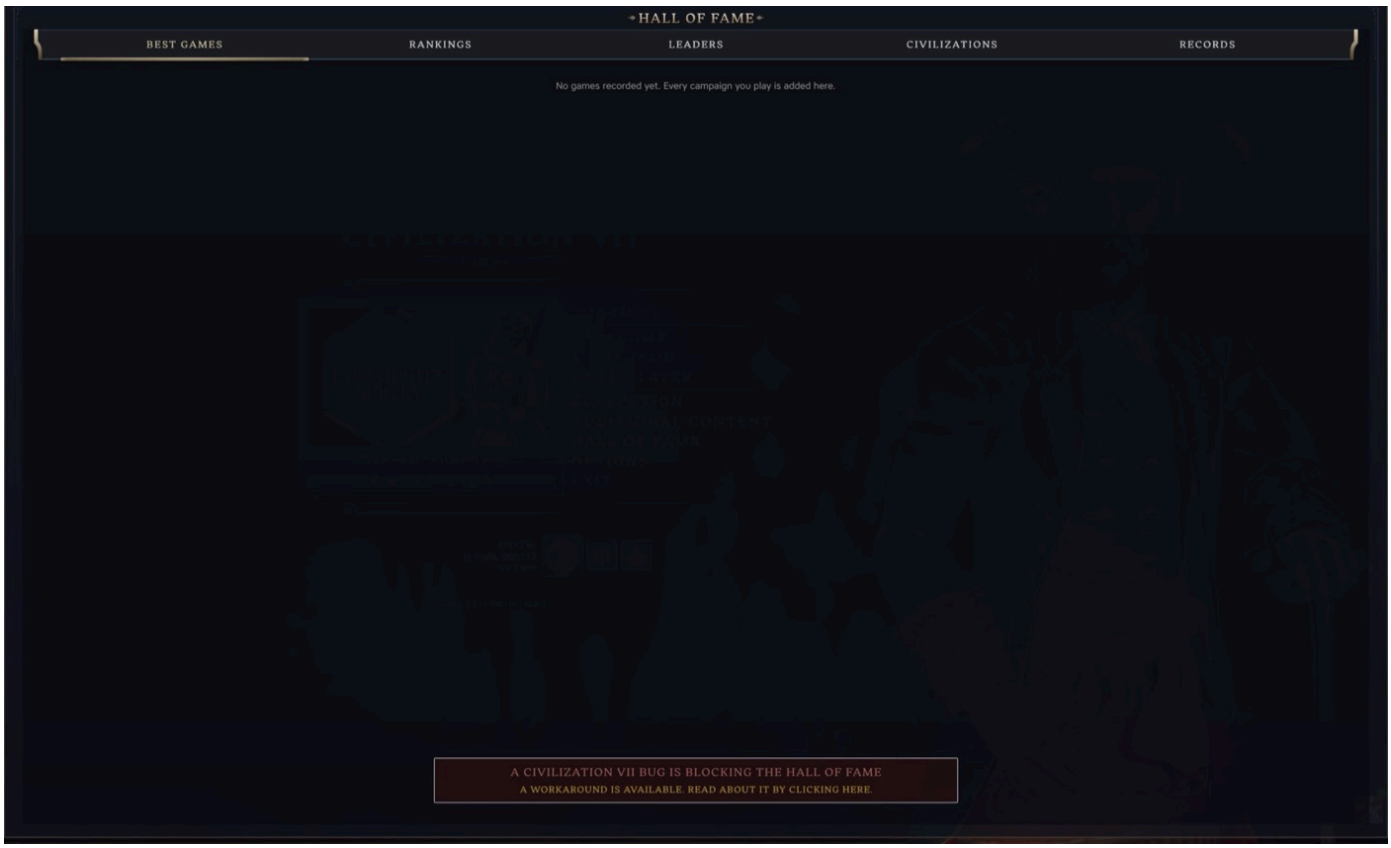

Figure 36: Repair mod storage: read first, then two clicks

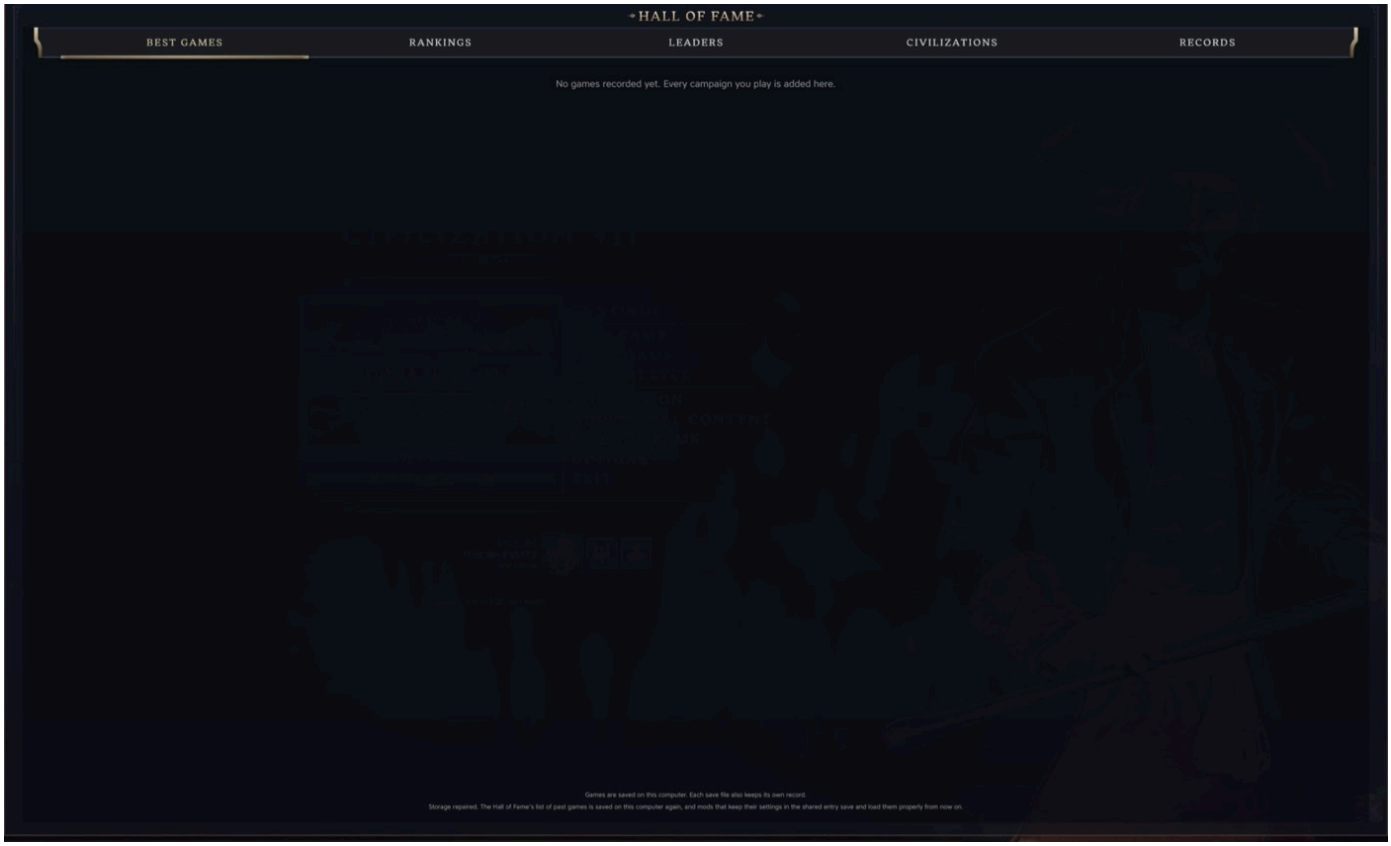

Figure 37: After the repair

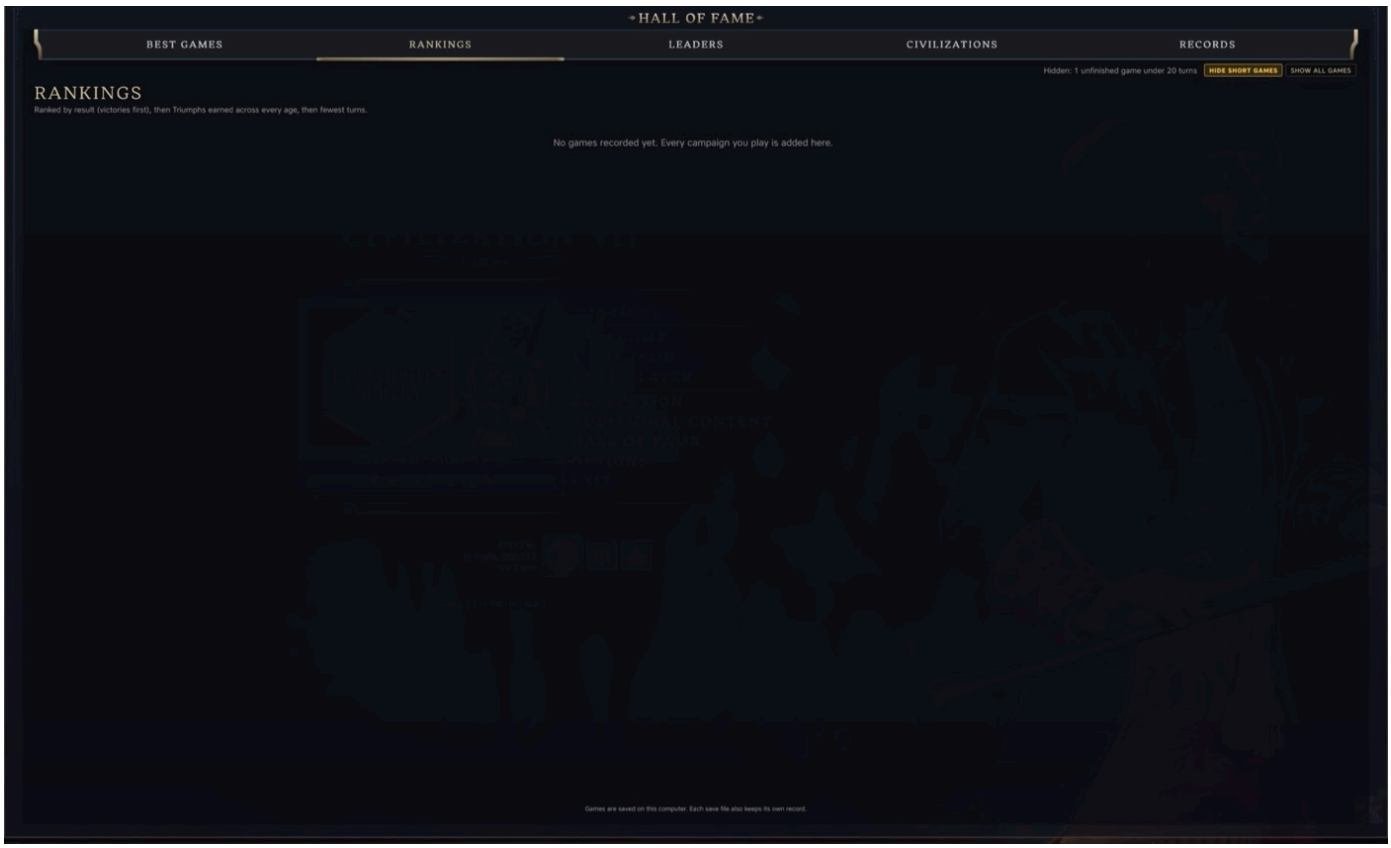

Figure 38: Hide short games or show all, on every page, with a count of what is hidden

Cities, or **Towns**.

- **Hall of Fame:** every campaign played on this computer, the current one included live. **Best Games** opens first: your totals, a podium of your three best games, the rest of your top ten, and the game you are playing at its rank between the games just above and below it (from the main menu, your most recent game), with victories by type. **Rankings** orders the games by result (victories first), then Triumphs earned across every age, then fewest turns, and gives each an honorific from Augustus Caesar down to Ethelred the Unready. **Leaders** and **Civilizations** give each one a win rate over finished games, drawn as a bar that is green for the games won and red for the rest, with both parts named, and the attempts, most Triumphs and average length beside it. **Records** names the game that holds each record (most Triumphs, most wonders, largest empire, most settlements taken, fastest victory, longest game). Any game opens a page with its lineage and figures, the territory map beside its rivals, the game's timeline (with playback), and its highlights, age by age, where each Triumph says what it was earned for and what it gave. Each page carries its name and a line saying what is on it. **Game options** at the foot of that page can remove the game from the Hall of Fame. Short unfinished games are hidden unless **Show all games** is picked. The **Hall of Fame** button on the main menu opens the same view with no game loaded. **Hide short games / Show all games** sits under the section tabs on every page: unfinished games under 20 turns are hidden by default as test loads, and the row says how many it is hiding. When a Civilization VII storage bug keeps the list from being read or saved, a notice box on every Hall of Fame page names the bug, and clicking it opens **Repair mod storage** (see Compatibility).

Across the first four, places 1–3 carry a gold, silver, or bronze wash, your own civilization and settlements are outlined in gold, and in the two tables the cell of whoever leads each column is highlighted in gold (hover it for the category). Civilizations you have not met show as placeholders unless name revelation is enabled in Options.

6. Conflicts and crises

The **War Timeline** is a Gantt chart of wars from diplomacy events (declarer, supporters, opposers, and a stable war ID), each with a generated name. **War Impact** picks one war and charts each combatant's trajectory through it: military strength, units lost, production directed to war, settlements and land lost or gained, settlements razed, population lost, and more. War cost is the observed change in each side's figures across the war, derived from the recorded samples (not invented).

Crisis Impact breaks each age's crisis into its stages (Begins, Intensifies, Culminates, Ends), each with a per-civ cost table, then a per-crisis cumulative-impact table and, once crises span more than one age, a cross-age overall total. Because the loss figures are sums of per-turn declines (which need dense samples), each finished age's cumulative cost is captured when that age ends, so it stays accurate even after old samples are thinned to cap the save.

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7. How the figures are calculated

Every number is **derived from what the mod can observe by sampling the game each turn**, not read from a hidden engine ledger. Each turn it records a snapshot of every met civilization's raw figures; the charts and tables compute everything from those snapshots. Two consequences follow: a figure can only be as fine-grained as the sampling (see §8), and "loss"-type figures are inferred from how a value moved over time rather than from an authoritative casualty log.

Direct reads

Most metrics are read straight off the player each turn with no transform: the per-turn yields (Gold/Production/Science/Culture/Influence/Happiness per turn, net), and the counts (Settlements, Techs, Civics, Wonders, Trade Routes, Ongoing Deals). These plot as-is.

Derived and rescaled figures

A few figures apply a deterministic transform so the raw game number reads at a believable real-world scale. These are **cosmetic** and never affect gameplay:

- **Population (scaled):** derived from **Civilization VII's own per-era growth formula** (the food cost the game charges to grow a settlement, which differs by age), turned into a representative people count by a single calibration constant. Each settlement is valued for *whatever age it's in* — towns in the thousands, great cities up

to ~1M pre-modern, and 10–38M megacities only in the Modern age — with a smooth, continuous hand-off across age boundaries (no jump at an age change) and an upper safety bound. A small per-settlement variation, drawn from real signals (happiness, urban/rural mix, growth trend), keeps any two same-size settlements from reading identically. The **civ-wide** figure is the **sum of its settlements'** estimates (not a separate, hotter aggregate), so "empire" and "sum of cities" agree. (The Emigration companion mod shares this exact curve, pinned by a cross-mod test.)

- **GDP:** a weighted sum of the per-turn yields, `× turn × 1,000,000`, shown in `$M`/`$B`. Weights: Gold 1.0, Production 1.0, Food 0.5, Science 1.2, Culture 1.2, Influence 1.5. Multiplying by the turn count approximates a cumulative economy rather than a single turn's output; the turn factor is capped at 300 so very long games and slow speeds keep a comparable scale.
- **Score:** `techs + civics + 2 × settlements + gold ÷ 100` , with techs and civics counted cumulatively across ages so the line never drops at an age change. The **Power Race** plots each civilization's rank by this score.
- **Land Area:** `owned_tiles × 7,000 km²` (a hex's nominal real-world area).
- **Military Power:** the summed combat strength of the civ's military units, totaled in the sampler (there is no clean player-level engine accessor).
- **Diplomatic Approval:** a reputation aggregate. Each met major civ contributes by relationship (Allied +5, Helpful +3, Friendly +2, Neutral 0, Unfriendly -2, Hostile -3, At War -5); suzerained city-states contribute the same weights at `× 0.3`.
- **Settlement Cap Utilization:** `settlements ÷ settlement_cap × 100`.
- **Land Area Share** and **Population Share:** each civilization's land (or population) as a share of the total held by all charted civilizations that turn, stacked to 100%.
- **Conquest %:** settlements a civilization has conquered, as a percentage of the settlements it holds now.
- **Crisis Stage:** the engine's internal stage (pre-crisis -1 through 3) shifted up by one so it reads "Stage 1" to "Stage 4" and plots cleanly as a step.

War and crisis cost

War costs (the per-combatant tables) and crisis costs (the per-civ stage and cumulative tables) share one engine: each figure is a participant's metric series reduced over its **active window** (`[join turn, leave turn or war/age end]`) by a mode chosen per figure:

- **Losses** (Strength Lost, Population/Crop/Production Lost): the **sum of every turn-over-turn decline** in the series, with rises ignored so ordinary growth can never mask a loss. This needs reasonably dense samples to catch each dip (see the caveat below). Population Lost reads **raw** population, not the rescaled chart value, so neither the people-scaling nor the age-reset bridge on the Population chart can hide real drops.
- **Net** (Settlements, Land): `last - first` over the window, signed (`+gained` / `-lost`). These are event-based (cities/territory that actually changed hands by capture), so they read "-" for wars that predate the tracking rather than a misleading 0.
- **Accrued / Spent** (Settlements Razed, Production Directed to War, Refugees): the increase of a cumulative event counter over the window (`last - first`).
- **Level** (current Military Power): the last sampled value, a standing figure rather than a flow.
- **Casualties** (Military Strength Lost) prefer the engine's cumulative units-killed counter when present (true kills), falling back to the standing-army decline for older saves.

A crisis's cumulative impact sums these across its stages, and a finished age's cumulative is **snapshotted when that age ends** so it survives later sample thinning (§6).

Caveat: sampling resolution

Because losses are summed from per-turn dips, coarse sampling (or the decimation of old samples that caps long games, §8) can **under-count** a loss, and on a heavily thinned window a loss figure drops to "-" rather than guess. Standing figures (current Military Power, counts) survive on a single sample. This is why the loss math reads raw, ignores rises, and why crisis cumulatives are snapshotted at their age boundary.

8. Behavior and persistence

- Settings persist in `localStorage`. Recorded history persists per save game via the GameConfiguration store, carrying across quit/load and age transitions. The chronicle is saved the same way, in the save file.

- The Hall of Fame keeps one compact record per campaign in the shared `modSettings` entry of `localStorage`, beside the settings, so it can be read from the main menu. It writes only when that entry holds settings it recognizes and leaves every other mod's settings as they are. Each save also carries its own campaign's record, which is added back to the Hall of Fame whenever that save is played. Names are stored once for the whole archive rather than inside each game, so a game with its whole story is small and a full Hall of Fame keeps every game's timeline. Territory maps are the largest part, so they are kept for the most recently played games; past that, the older games give up their maps, then the other civilizations' timeline tracks, then their timelines, before any game is dropped; the game being played is always kept whole. Recording the same game again (a save from before Demographics was added, loaded twice) updates its one entry rather than adding another. When the Hall of Fame's list cannot be read, a note under it says so, names the game bug, and offers to repair the shared entry, which deletes one other mod's saved data (see Compatibility).
- History sample caps are **automatic**: they scale with game speed and adapt to hardware (CPU cores / device memory / mobile) and game size (player count), shrinking retention on weak machines and many-civ games (floored so history is never starved). Chart rendering is additionally clamped to a per-series point budget scaled by the same capability factor, so a marathon-length line plots a bounded number of points after the visible-range filter.
- **Complexity** (game Options → Mods → Demographics) sets how much of the dashboard is shown: **Basic** shows the core pages (Yields Per Turn, Economy & Resources, Society, Land & Settlements, Population, Global Relations, Military Power); **Standard** (default) and **Analyst** show every page, adding Religion, Agreements, Crises, and Soft Power.
- Colorblind mode swaps chart and relationship colors to a colorblind-safe set.
- **Spoiler guard** (on by default, Options → Mods → Demographics) hides the names, charts, and diplomacy and relations stats of civilizations you haven't met, so unmet civilizations render as placeholders and their lines show a gap. **History on first contact** chooses whether meeting a civilization reveals its full back-history or tracks it only from first contact. A banner below the views states the active analytics policy.
- Multiplayer governance: an analytics policy stored in the shared `GameConfiguration` is enforced as a ceiling for every player that game — clients drop any civ the host's policy hides, and the banner notes when the host (rather than a local preference) is the binding constraint.

9. Companion-mod integration

Other mods can contribute to the dashboard through an optional, order-independent API on `globalThis.DemographicsMetricsAPI` (inert unless called, so the base mod is unchanged):

- **`registerMetric(spec)`**: add a per-civ line-chart metric that flows through the normal sample, store, and chart pipeline. A spec may include a `tooltipAttribution(ctx)` callback whose returned string is shown as a source-attribution line in that metric's tooltip.
- **`registerMetricToPage(pageId, metricId, afterMetricId?)`**: place the metric's tab on an existing page (for example, next to Population).
- **`registerMetricGroup(spec)`**: collapse several related metrics behind one tab with up to two pill-row toggles (a metric row and a view/units row), charting one member at a time. Member metrics stay individually registered (and sampled); the group just picks which one is shown.
- **`registerPanel(spec)`**: contribute a whole **page** (optionally with its own sub-tabs) whose body the companion renders itself. The screen adds it as its own top-level tab and calls `spec.render(container, ctx, subId)` to fill it (the chart title, time-range filter, and CSV toolbar are suppressed for these custom pages). Demographics gains no dependency on the companion.

The handshake is load-order-independent: registrations made before this screen loads are queued and drained when it initializes. The companion **Emigration** mod uses all of these, contributing an **Emigration** tab whose **Population & Migration** page charts Population, Population Share, Net Migration, Emigration, Immigration, and Refugees in either scaled "people" or raw Civ population numbers, with cause-breakdown tooltips and war/disaster event markers — alongside its own Network, Causes, My Cities, Diversity, Policies, Notifications, and Guide pages.

Performance on large saves (with Emigration installed)

If late-game turns feel heavy with both mods active, **raise Emigration's *How often migration runs*** (game Options → Mods → Emigration → Advanced, Pacing group) so its migration pass runs less often. This is a pure cadence lever: it changes how often migration updates, not behavior or graph semantics. (Demographics' own sampling cadence is automatic and adapts to hardware/game size, so there's no manual frequency knob to tune.)

Measuring the combined cost (developer recipe, not gameplay): on a turn where both mods fire, the debug logs report Emigration's pass duration and Demographics' sample duration; opening the **Emigration** tab exercises the shared render core. Comparing those three tells you whether a turn spike is the Emigration pass, the Demographics sample, or the embedded page: the cross-mod bridge itself is a thin read-only layer over each mod's existing tallies.

10. Install and run

1. Subscribe, or place the `demographics` folder in the Mods directory:
   - Windows: `%localappdata%/Firaxis Games/Sid Meier's Civilization VII/Mods/`
   - macOS: `~/Library/Application Support/Civilization VII/Mods/`
2. Enable Demographics in Additional Content.

11. Usage

- Open from the Demographics button in the subsystem dock, the pause menu, or the end-of-game screen. From the end-of-game screen it opens on the Hall of Fame. The **Hall of Fame** button on the main menu shows past campaigns with no game loaded.
- The top tabs pick a hub (Global Statistics, Geopolitics, History, World Rankings, plus **Emigration** when that companion mod is installed), the second row picks a page, and the pill row picks the chart.
- The **Options** button (in the chart toolbar, and at the top right of Global Relations and World Rankings) opens the game's Options screen at **Mods → Demographics**.

12. Compatibility

- Does not overwrite base-game files.
- All engine reads are defensive: schema drift yields missing data, not crashes.
- Persistence uses the GameConfiguration store (`Configuration.editGame()` / `getGame()`). If that API is unavailable, history is kept in memory for the session only.
- **A Civilization VII bug, and the repair for it:** Civilization VII 1.5.0 has a bug in the storage mods use for their settings: whenever a mod asks for its own saved data, the game returns the saved data of whichever mod's entry sorts first. Only that one mod's data works; the entry that Demographics and most mod option panels share can never be read back, so other mods' settings reset every launch and the Hall of Fame cannot read or save its list of past games (every game is still stored in its own save). It is a game bug for Firaxis to fix; a reproduction is written up for their support portal. When it is blocking the Hall of Fame, a notice box on every Hall of Fame page names the bug, and clicking it opens **Repair mod storage**: it empties the storage and starts it again with one entry, the shared settings entry most mod option panels use, holding this mod's settings and past games; once it is the only entry it is what the game returns, so every mod that keeps its settings there loads and saves them normally from then on, including after a restart (a mod with a separate entry of its own is not helped: it could not read it before and still cannot). It permanently deletes the saved data of the one mod the game was reading for everyone, and clears what other mods had written but could never read back; saved games, the history inside them, the game's own settings and every mod's file are untouched. The sheet says all of this before the button, the button says what it deletes, it takes two clicks with a Cancel, and it is offered again if a mod writes its own first-sorting entry later. The whole story: docs/civ7-storage-bug.md.

Credits & license

- robk: Demographics (Civ V).
- Gedemon: CivGraphs (Civ VI).
- Slothoth: Global Relations (Civ VII).
- The Civilization modding community for documentation, samples, and testing.
- Tower: Civilization VII rebuild; History and Hall of Fame carried over from Tower's History & Rankings mod.

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